



TG1001 - Batch Processing - File Reconciliation

PowerCARD-Solution

Client: Product
PowerCARD version: V3.5
Document version: V1.1

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1 Introduction

1.1 Document objective

This document is an operational guide of PowerCARD solution. It is useful to understand the processes implemented within the solution to allow the customer to define their internal operational procedures.

1.2 Operating context of the solution

Skills required for managing PowerCARD solutions from operating point of view are Oracle DBA and a system engineer (*AIX and Websphere*).

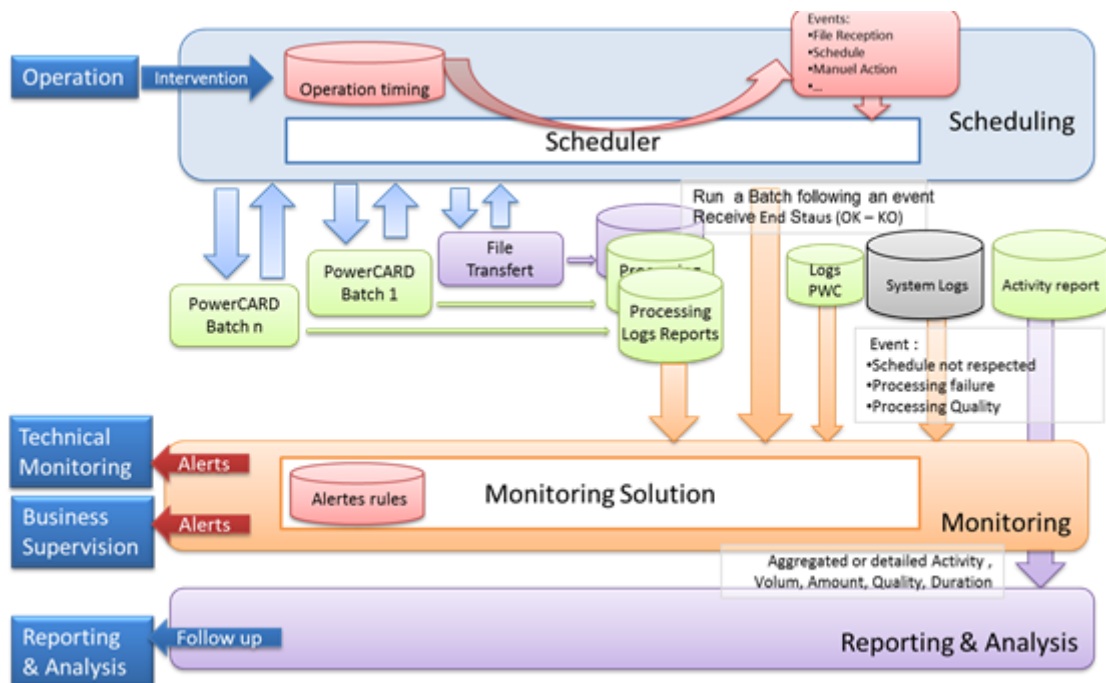
Common administration tasks are:

1. Evaluate the database and application server hardware
2. PowerCARD Application and database backup
3. Monitoring data volume,
4. User management,
5. Batch organization and monitoring.

1.3 Operations principles

The figure below depicts the main technical actors handling PowerCARD operational environment:

Figure 1: Operation Administration and Management



In general, the operation administration and management covers the activities:

1. Technical operations,
2. Technical monitoring,
3. Business supervision,

4. Reporting and analysis.

Hereafter the definitions of main flows identified in the application and which are helpful to understand the next sections of the document:

1. Return code of a batch to indicate if the process ended successfully or unsuccessfully. This information is communicated to the batch scheduler (ex.: Dollar Universe, Control M...),
2. Return flow as a response to a processing flow : it is communicated to the third system which has initially initiate the process, to advise it about what has been performed (ex. : CFT acknowledgment, return file ...),
3. Event report « CRE » or activity report: It is a file used for reporting and analysis further to the extraction of data from the database,
4. Processing report: information stored in log files communicated to the supervision agent like PATROL.

Four levels are identified to manage the day to day operations and the incidents:

- Level 1 : technical operations,
- Level 2 : system infrastructure administration, software's, production network,
- Level 3 : technical-functional support of PowerCARD application,
- Level 4 : HPS support.

2 Batches processing

In this section, we describe:

1. The definition and architecture of a PowerCARD batch
2. The diagram syntax of a PowerCARD batch
3. Architecture batches
4. Running a batch
5. Encryption

Batches will be designed by (*batch_name*) within this section.

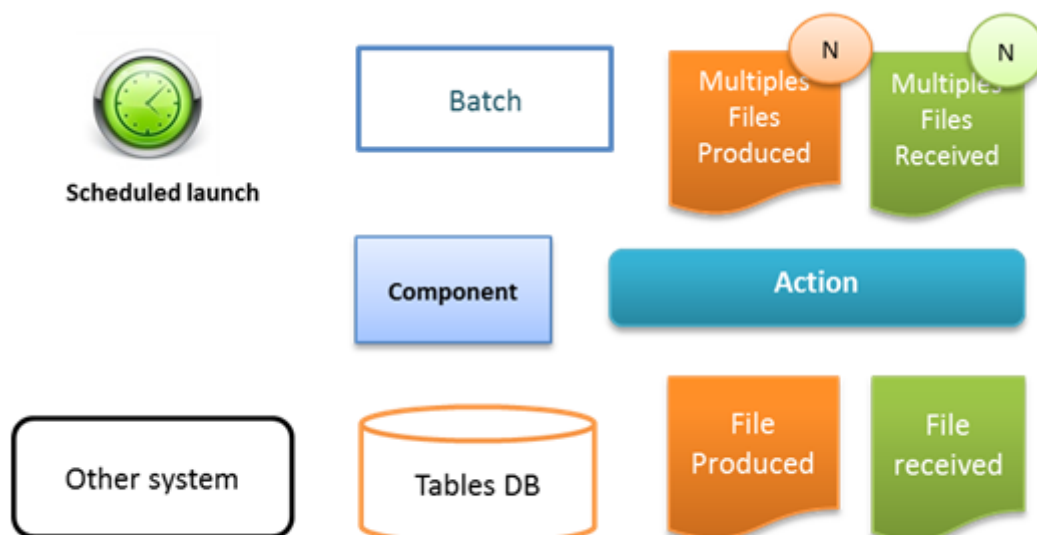
2.1 Definitions

A PowerCARD batch is a set of processes that is using the data available in the database to perform functional processes, generate files or load files.

A chain refers to running a group of these kinds of batches.

2.2 Diagram Syntax

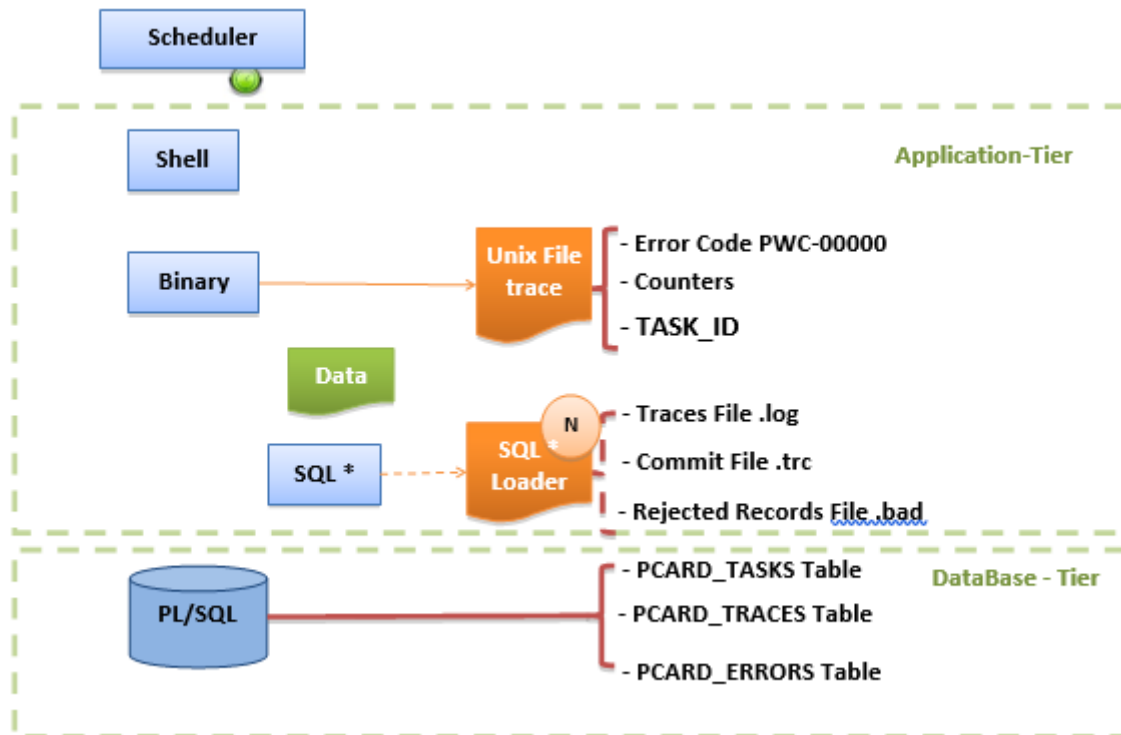
Figure 2: Batch "figures syntax"



2.3 Architecture

In 3 tiers architecture, batch components are shared between application tier and database tier. PowerCARD batches (*standard & ETL*) follow the same rules for setting inputs parameters and outputs trace file.

Figure 3: Architecture Components



2.3.1 Components of the application tier

2.3.1.1 Shell

It is the starting point of a process. Two parameters are communicated to the shell:

- Names of the binary,
- Log file to use.

2.3.1.2 Binary file

It is the key driver of the batch:

- Get the parameters defined at the task level
- Load, if needed, the input file(s) using the control files
- Call the PL/SQL functional process
- Generate output files
- Store the summary of the process in its own log file
- Send back the execution error code [1: OK, 11: Warning and 99: Error].

2.3.1.3 Ctl file

The SQL*Loader control file is a text file that contains data definition language [DDL] instructions. The DDL is used to control:

- Where SQL*Loader will find the data to load
- How SQL*Loader expects that data to be formatted
- How SQL*Loader will be configured (*memory management, rejecting records, interrupted load handling, and so on*) as it loads the data
- How SQL*Loader will manipulate the data being loaded

Control file will be present only for batches which need to load data from file into the database.

The processing is performed on two steps. The first one is loading the content of the file in a temporary table(s). The second one refers to the functional process where the data is extracted from temporary table(s), processed and stored in the destination tables. If an error occurs in the first step the temporary table(s) are purged and the process need to be performed from the beginning (*total rollback*).

**Note**

This file is only created when the process updates tables based on the input of a file.

2.3.1.4 Grf file

It refers to the configuration graphs files used by the ETL module. The graphs files are called directly from the shell once the ETL is launched.

2.3.2 Component of the database tiers

2.3.2.1 Task in Pcard_Tasks

A task or a batch is defined in PCARD_TASKS table and a new task can be parametrized in this table.

Generally, there is one entry per task in PCARD_TASKS table. This entry contains the processing date and the parameters required by the binary file. The task has usually the same name as the binary with some exceptions.

To handle performance issues for batches that deal with large amount of data, a parallel execution concept is used by PowerCARD. The parallel process uses two parameters tables PCARD_TASKS_JOBS and PCARD_TASKS_JOB_PARAM to set the number of simultaneous jobs and other required parametrizations.

2.3.2.2 Packages related to a task process

The functional process of a task is contained in PLS/SQL packages. Once the parameters are communicated to the main package and eventually data is loaded in temporary tables, the functional process may start.

Error messages are logged in PCARD_TRACES and PCARD_ERRORS, while the processing status, OK or KO, is transmitted to the binary file to be written in its trace file.

2.4 Running a batch

In the application tier, we are using environment variables to access rapidly to the working directories:

- \$BIN: directory for binary files
- \$SHL : directory containing shell scripts
- \$CTL : directory for SQL*Loader control files
- \$TRACE : directory which contains Trace files
- \$DATA: directory where received and generated files are stored. Note that we can manage storing files under sub directories of \$DATA directory.
- \$ARCHIVE : directory where files are moved to

To start a batch, 2 steps are required:

1. Set in the database the batch parameters
2. Launch the shell file from the application tier

2.4.1 Defining the batch parameters

The parameters of a batch (task) are defined in PCARD_TASKS table; they need to be filled or updated each time the batch is run:

- **STATUS:** To be set to **P** (*Pending*)
- **BUSINESS_DATE:** Date of the processing.
- **PARAMETER_V1_DATA:** 1st parameter of the task (*depends on the task*)
- ...
- **PARAMETER_V10_DATA:** 10th parameter of the task (*can be null it depends on the task*).

Screen path: **Administration > Jobs > Jobs parameters**

Figure 4: Jobs parameters (Details screen)

Figure 5: Jobs parameters (Parameters tab)

If the batch is executed from the screen, the user is required to capture the batch parameters, and the update of the database is done automatically.

The screen above is defined during the initial deployment of PowerCARD, and it is not recommended to modify these settings. Since these settings will only affect the launching screen and not the batch itself, if a parameter is set to not be mandatory, this parameter remains mandatory while the screen would not display it, and eventually create incoherences.

If the batch is executed by a script used to automate the batch scheduling, the following script \$BIN/update_pcard_tasks is required to update the Bank Code, task parameters, task status and the business date.

Syntax to call the "update_pcard_tasks" binary file:

```
$ $BIN/update_pcard_tasks update_pcard_tasks <log file update_pcard_tasks>
<task_name> <bank_code 123456 <business_date_DDMYYYY> <PAR_1> <PAR_2> <PAR_3>
<PAR_4> <PAR_5> <PAR_6> <PAR_7> <PAR_8> <PAR_9> <PAR_10>
```

Where:

- **Bank_code:** The institution ID number to have this task executed. This field is mandatory.. Several banks can be included separated by commas.
- **Task_name:** Name of the task to be executed.
- **Business_date_DDMYYYY:** business date to communicate to the batch. It has to have BDDMMYYYY format, or " to use the current business date of the system.
- **PAR_X:** Value of the X parameter or Null. 256 characters maximum.

The batch returns 1 (OK) or 99 (KO) and does not generate neither a log file nor an error code. It is not considered as PowerCARD batch.

Return code listing: 00004, 00019, 00021, 99999.

2.4.2 Most common parameters examples and values

PowerCARD provide maximum flexibility in the scheduling of the batch by reusing the same set of parameters for several batches. Here is a short listing of the most common parameters with their possible values, or since the parameters are configuration based how to obtain them:

- **Origin code:** Origin code value is to design the source and type of the transaction. Possible values are:

Source description	Type	Possible values
ON US CARD / LOCAL MERCHANT	CHAR(1)	0
ON US CARD / LOCAL OTHER BANK MERCHANT	CHAR(1)	1
ON US CARD / FOREIGN MERCHANT	CHAR(1)	2
LOCAL OTHER BANK CARD / LOCAL MERCHANT	CHAR(1)	3
FOREIGN CARD / LOCAL MERCHANT	CHAR(1)	4

- **Network code:** This is used to identify the network either as a source or destination of the transactions. These are the typical values:

Source description	Type	Possible values
VISA	CHAR(2)	01
MASTERCARD	CHAR(2)	02
EUROPAY	CHAR(2)	03
AMEX	CHAR(2)	04
DINERS	CHAR(2)	05
JCB	CHAR(2)	06
DISCOVER	CHAR(2)	07

**Note**

Other values may be added for private network

- Bank code: Bank ID:** This is the identity code of Institutions managed by the PowerCARD platform. As this information varies per customers, it is not provided in this document. However, a PowerCARD user may get the listing via **Parameters > Institution > Institution** or via a database query : `'select bank_code, center_code, bank_name from BANK;'`.
 Furthermore, this parameter can be used in the following fashion:
 - Specific bank code: "123456" the batch will be executed for this specific Bank ID
 - For all Banks
 - Non reporting Batch: the parameter 'ALL' must be specified to process data for all Institutions defined in Platform
 - Reporting batch the parameter is null to generate reports or data files for all Institutions defined in Platform
 - For a discrete number of Banks: separated by commas: "123456, 234567" this mode of operation only works for reports i.e. Reporting Section.
- Reason code:** This is listing of reason code for transactions.
 Possible values are:
 NON PAYMENT OF FEES, SIGNATURE DIFFERS, Reverse TC - without fee etc. As these values vary per customer, a full listing is not provided. However, a PowerCARD analyst may access the full listing via the screen : **Parameters > Platform > Transaction codes** or via a database query : `'select bank_code, reason_code, reason_code_wording from TRANS_REASON_CODE;'`
- Card product:** Describe the type of product (*VISA DEBIT, MC Platinum Credit Card, MC Gold, etc.*). As this is specific per customer, it is not described in this document. However, an analyst with access to PowerCARD can obtain a listing via the screens **PARAMETER>GENERAL>CARDHOLDER:PRODUCT CODE** or using database access via the query : `'select bank_code, product_code, wording from CARD_PRODUCT;'`
- Transaction definition:** Describe the type of transaction. As this is specific per customer, they are not specified in this document. However, a full listing may be obtained either via the PowerCARD screen : **PARAMETER>ISSUING>REVOLVING CREDIT:TRANSACTIONS:TRANSACTION REASON** or via a database query: `'select transaction_code, wording from TRANSACTION_DEFINITION;'`
- Account profile code:** This is used to describe the type of the account (debit, pre-paid, credit, etc.). As this is specific per customer, these are not detailed in this document. However, an PowerCARD analyst may obtain them via **PARAMETER>ISSUING> REVOLVING CREDIT>PROFILE:CREDIT PROFILE DEFINITION:** or via a database query: `' select bank_code, profile_code , wording from CR_PROFILE;'`
- Language:** User language is selected via the screen at the login stage or specified via parameter during command line execution. The most common values are en_EN for English, es_ES for Spanish and fr_FR for French.
- Account status:** Administrative status of the account (Normal, Collection, etc). These are specific per customer and are not detailed in this document; however, here is how to obtain the full listing via PowerCARD **PARAMETER>ISSUING>REVOLVING CREDIT:CREDIT ACCOUNT STATUS** or via a database query : `'select profile_code, wording from SHADOW_ACCOUNT_STATUS;'`

2.4.3 Update of the business date

If the business date is left to NULL, the current business set in PowerCARD will be used.

The choice to pass the business date is left up to the operational team; however it is a best practice to systematically include the business date as part of the update of the script to avoid errors.

The actual value of the business date is an operational decision or based upon business requirement.

2.4.4 Execute a shell

From the application tier, the shell file is executed and the trace file name to use is mentioned. The name of the trace file padded by 000, and this number is incremented by 1 each time the process cannot write in the log file.

Example: `load_posting_new_file.sh load_posting_new_file.trc000 POSTFLIN` will generate a trace file `load_posting_new_file.trc000` and if the trace file reached the maximum size "1024" bytes, the shell will increment the sequence and generate `load_posting_new_file.trc001` file.

2.4.5 Batch description model

The following section describes the common batches between the Acquirer and Issuer products.

The description is structured as follows:

- Batch flow,
- Batch description,
- Batch components,
- Batch parameters,
- Parameters example,
- Input data,
- Output data,
- Counters,
- Warning,
- Recovery procedure,.

2.4.5.1 Batch Flow

The flow contains a diagram description of the batch with a list of the main interactions and the input/output data.

2.4.5.2 Batch Description

It is the functional description of the batch and we mention execution constraints if any.

2.4.5.3 Batch components

The table below contains name, paths and physical component of the batch (*Shell, binary, task name, log file ...*)

Shell	Name of the shell to run the batch
Binary	Binary file name and directory where it stored(binary is called by the shell file)

Control file	Control file name and directory where it is stored. Used for batches having a file as input.
Task	Task name (<i>batch name</i>)PCARD_TASKS.TASK_NAME where parameters are set.
Trace file	Trace file name and path of the directory where the file is generated
Command	Shell command to run the batch either from unix/Linux or from a scheduler

2.4.5.4 Batch parameters

It refers to the task input parameters description; these are set at the beginning before the functional process start. There is up to 10 parameters PARAMETER_VX_DATA in table PCARD_TASKS in addition to the BUSINESS_DATE which is the processing date.

Column name	Description
BUSINESS_DATE	All batches can set the business date
PARAMETER_VX_DATA	Xth parameter to set

Below is the table layout that will be used to describe the meaning of each parameter and specify whether it is mandatory or not:

Column name	Parameter name to set
Parameter	Label/meaning of the parameter
Description	Usage description
Mandat.	Mandatory parameter

2.4.5.5 Parameters examples

It is the description of valid parameters combinations according to the use case of the batch. It is a summary of the batches uses cases versus parameters values.

How to read the table:

Case	Id of the use case
Description	Functional description of the case
Chain/recovery	To indicate if the batch is part of a chain processing or it is a recovery batch or it is out of chain.
Parent	To indicate the batch and the parent id case if both batches are linked.
Child	Recommended batch to be run next if relevant
PARAM_VX	To provide values of the batch parameters 1..10

2.4.5.6 Input data

Files have to be in \$DATA:

- If the batch ended with OK or WARNING: the file is moved to \$ARCHIVE and has the suffix timestamp .DDMMYYHH24MI
- If the batch ended with ERROR: the file remains in \$DATA

The description of the input data is provided in the next chapter.

2.4.5.7 Output data

Generally the output files are generated under \$DATA directory

- If the batch ended with OK or WARNING: the file is copied to \$ARCHIVE and has timestamp suffix .DDMMYYHH24MI (+ other information depending on the batches)
 - If the batch ended with ERROR: the file is either not generated or it remains in \$DATA
- Reject files/remittance reports (RR) are generated under \$DATA
- If the processing ended OK: the file (empty) is copied under \$ARCHIVE and has suffix timestamp .DDMMYYHH24MI
 - If the processing ended with WARNING or ERROR: the file is copied under \$ARCHIVE and has suffix timestamp .DDMMYYHH24MI

The input, output elements (files, tables...) of a batch are described in a table with 3 columns like the below one :

Type	Element type input/output. Table file
Name	Name and location of the element
Description	Functional description of the element

These parameters are describes in a table which can be read as follow:

Name	File name
Type	Variable with separator, record fixed length, binary EBCDIC
Size	Maximum lengths of records
Format	File format

2.4.5.8 Counters

It contains the description of the counters that will be written in the batch log file. It is an extraction of the text part only of the log file, the prefix part plus timestamp and PIDs are removed.

These counters have to be analyzed in day to day operations. Classic counters are:

- Processed (*Total number of records*)
- Succeed (*number of records successfully processed*)
- Rejected (*number of records unsuccessfully processed*)

A record can be an authorization, a transaction...

2.4.5.9 Warning

This paragraph aims to describe if a process can end with Warning. The batches can end with Warning when a subset of records cannot be processed for technical reasons and not functional ones. Some batches have a threshold parametrized in the table REMITTANCE_AGENT so that the batch will end with ERROR if it's met. The other batches keep WARNING status.

Relevant SQL queries will be provided as well as the relevant error codes for the specific batch.

2.4.5.10 Recovery procedure

This chapter has to be used when there is an error during the execution of a batch and once the origin of the error is identified and corrected.

It explains the recovery procedure mode, the parameters to set to re-start the batch and if the batch will start the processing from the beginning or from the crash point. It is mentioned also if the chain has to be stopped and the database needs to be cleaned before restarting the batch.

The information is summarized in a table like this one:

Blocking	To indicate if further to the batch failure, the chain has to be stopped.
Clean	To indicate if some data need to be cleaned before re-run of the batch.
Mode	Recovery mode: - Not available – no recovery - From the beginning – The process restart from the beginning - Crash – recovery, starts from where the process has stopped. A batch parameter has to be set in this case
Batch	Batch name from where the recovery procedure need to start.
Parameters	Use case, of this process, to be performed to start the recovery procedure.

The recovery procedure and recommended actions are further details into Ref001.

2.4.6 Batch multi-cutoff support

Multi-cutoff support has been introduced for some batches for the PowerCARD V3.5.

This functionality allows processor or large institution to run concurrent jobs in parallel for a pre-configured group of institutions.

The institution groups can be defined in the below screen.

Screen path: **Administrations > Parameters > Jobs > Group bank jobs**

The screenshot displays the 'Group bank / jobs' configuration screen. It features a table with the following data:

Identifier	Name	Description
EPL000	DEFAULT GROUP	The default batch execution group

At the top, there are input fields for 'Identifier' and 'Name'. On the right, there are buttons for 'Search', 'New', and 'Clear'. The bottom of the screen shows a 'Columns' dropdown and a page number '10'.

Figure 6: Group bank/Jobs

For each **Group code** created, a directory with the **Name** of the created **Group code** must be created by DBA under the following paths:

- \$DATA
- \$ARCHIVE
- \$TRACE
- \$CTL

For the **group code** directory created under \$DATA and \$ARCHIVE, two other directories should be created:

- IN: To place/archive all the input files related to a **Group code** execution.
- OUT: To generate/archive all output files related to a **Group code** execution.

**Note**

The previous directories changes are required only when the UNIX environment variable "MULTI_CUTOFF" is set to 1 or **Group code** is different from DFLGRP.

For example: The execution is made with the group code 'DFLGRP'. The trace files will be available under:

- \$TRACE/DFLGRP/task_name.log
- \$TRACE/DFLGRP/task_name.trc
- \$TRACE/DFLGRP/task_name.bad

The data file should be placed under:

- \$DATA/DFLGRP/IN

2.5 Native file encryption

PowerCARD comes with native file encryption to comply with PCI requirements.

Implementation of the encryption is achieved via openssl libraries deployed during the initial setup of PowerCARD at the customer's premises. Customer must pay special attention to the choice of passphrase and its management. It is strongly recommended that this passphrase be managed as per Pin Security and Management procedures.

**Note**

No recovery of lost passphrase is possible. Please ensure dual control of passphrase are in place.

The exact type of encryption used is defined during the implementation yet these 4 elements will be produced:

- **Public key:** Used to encrypt all of the data. This file can be shared without issue. It must remain on the server to be able to encrypt new files.
- **Private key (in clear):** Used to decrypt the data. This file must be protected by a strong passphrase. The private key must not be stored on the server in clear. A copy should be kept in cold storage as part of the disaster recovery plan.
- **Private key (encrypted):** This file contains the secured version of the clear private key, and must be kept on the server to allow decryption.
- **Strong passphrase:** This is a more complex version of a password used to secure the clear private key. It will be required for every decryption files.

To decipher any files, the following command must be used :

```
openssl
inkey      smime -decrypt -in <encrypted filename> file -binary -inform DEM -
           $HOME/usr/cfg/rsa/ pwc_privatekey.pem      -out <Clear filename>
```

The user will then be prompted for the passphrase and the resulting clear filename be created in the \$data directory.

**Note**

- PowerCARD encrypts all the interfaces files exchange
- Report generated by PowerCARD must not be encrypted. As such a control file ensures that only the reports are kept in clear text.

The control file is located in: `$/SHL/encrypt_file.prm`

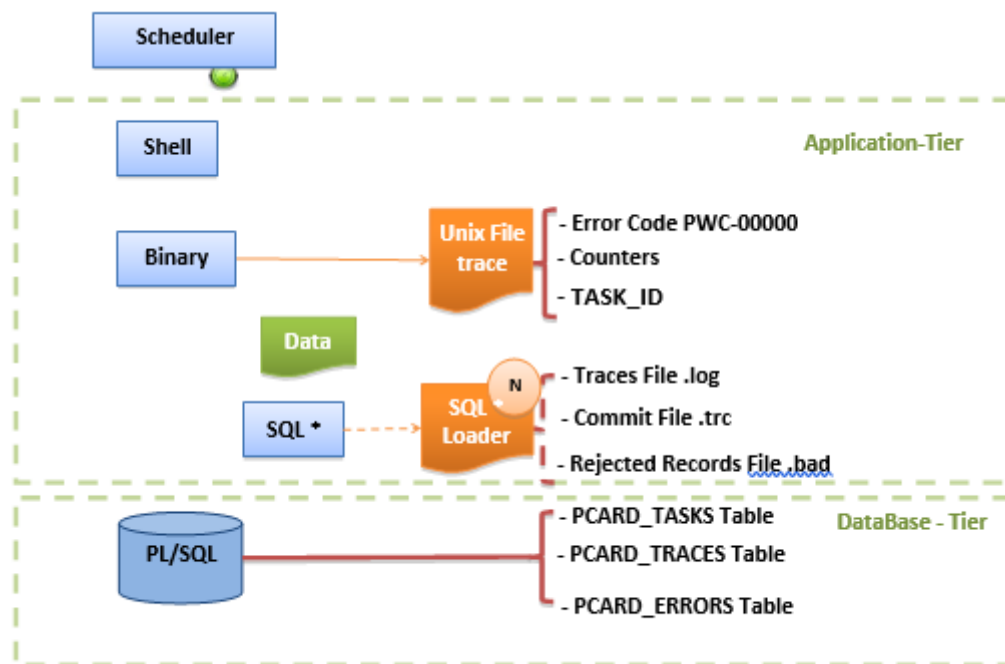
PCI considerations: the control file must be audited regularly and have the proper permission set to the right users:

- System administrator : full control and ownership
- PowerCARD users (*including batch users*): read only rights
- Other users: no access
- No execution privilege for anyone.

2.6 Exception management

2.6.1 Logs and traces architecture

Figure 7: Batch traces and logs



2.6.2 Batch processing logs and traces

Each job generates a trace file. The error codes are attached in Annex below. Possible Return codes are: OK, WARNING, KO.

Trace file

This file is located in the `$ TRACE` directory.

Structure

Each log line has the following logic:

- Date: Date of the trace line generation (*DDMM format*)
- § Space ' '
- § Time: Time of the trace line generation (*HH24MISSMMM format*)
- § Space ' '
- § PPID : Unix Process Parent Id
- § Space ' '
- § PID : Unix Process Id
- § Space ' '
- § TID : Unix Thread Id
- § Space ' '
- § '00001'
- § Pipe '|'
- § Trace Level : 0 (*HIGHEST*) to 5 (*LOWEST*)
- § Pipe '|'
- § Space ' '
- § Text: Content of the trace till the position 131, padded on the right with spaces.

Content

The log file of a given job is structured as follow:

- § The list of parameters used during the execution

2707 072700456 9895964 09634018 00001|0| (TASK NAME) [transaction_c_v3] .

```

2707      072700456 9895964 09634018 00001|0| (BUSINESS DATE) .....
          [27072012]
          .
2707      072700456 9895964 09634018 00001|0| (PARAMETER1  ) ..... [
          ]
          .
2707      072700456 9895964 09634018 00001|0| (PARAMETER2  ) ..... [
          ]
          .
2707      072700456 9895964 09634018 00001|0| (PARAMETER3  ) ..... [
          ]
          .
2707      072700456 9895964 09634018 00001|0| (PARAMETER4  ) ..... [
          ]
          .
2707      072700456 9895964 09634018 00001|0| (PARAMETER5  ) ..... [
          ]
          .
2707      072700456 9895964 09634018 00001|0| (PARAMETER6  ) ..... [
          ]
          .
2707      072700456 9895964 09634018 00001|0| (PARAMETER7  ) ..... [
          ]
          .
2707      072700456 9895964 09634018 00001|0| (PARAMETER8  ) ..... [
          ]
          .
2707      072700456 9895964 09634018 00001|0| (PARAMETER9  ) ..... [ ]
          .

```

```
2707      072700458 9895964 09634018 00001|0| (PARAMETER10 ) ..... [
      ]
      .
```

- The task Id

```
2707      072700458 9895964 09634018 00001|0| (TASK ID ) .....
      [11393]
      .
```

- The specific logs of the job
- The associated counters

```
2707      072740249 9895964 09634018 00001|0| -----Counter BATCH
transaction_ctrl_v3 ]
-----
2707      072740249 9895964 09634018 00001|0|
-----

2707      072740249 9895964 09634018 00001|0| ----- [ Base TRX :78 ]
-----
2707      072740249 9895964 09634018 00001|0| ----- [ RECYCL TRX :0 ] -----
-----

2707      072740249 9895964 09634018 00001|0| ----- [ VALID TRX :22 ]
-----
2707      072740249 9895964 09634018 00001|0| ----- [ REJ FONCT :56 ]
-----
2707      072740249 9895964 09634018 00001|0| ----- [ REJ TECH :0 ]
-----
2707      072740249 9895964 09634018 00001|0|
-----

      .
```

- The errors encountered during the execution of the job as well as the number of occurrences and the associated oracle error code if any

```
0204 114646687 13435038 19464240 00001|5| Start LogPcardErrors() .
0204 114646687 13435038 19464240 00001|5| ***** START PCARD_ERRORS
*****
0204 114646713 13435038 19464240 00001|0| --- ERROR : PWC-00004 count :
0000000002[ ORA-01403 ] .
0204 114646713 13435038 19464240 00001|5| ***** END PCARD_ERRORS
*****
0204 114646713 13435038 19464240 00001|5| END LogPcardErrors(OK) .
```

- The task execution status

```
2707 072740250 9895964 09634018 00001|0| ENDING STATUS TASK --> PWC-00000
```

The stat file

This file contains all display console or error of the shell program associated to the job, which would have occurred during the execution of the process. This log file is usually empty.

SQL * Loader traces

These are the standard log files of Oracle SQL * Loader.

- The files *. Log traces of loading showing the summary of processed records, lines with errors ...
- The *. Bad files:rejection files that contain the records that were not loaded
- The *. Trc files indicating when the intermediate “commits”were executed.

PCARD_TASKS Table

This is the table that contains all batches processing, their parameters and storing their execution status.

Content

Main columns of the table are as follows:

- TASK_NAME: It has the same name of the job
- STATUS: The batch status ('S'uccessful, 'U'nsuccessful ou 'P'ending)
- ENDING_STATUS: Return code of the last execution.
- BUSINESS_DATE : The processing date used for the job execution
- PARAMETER_VX_DATA : Values of the parameters X used for the job execution
- COUNTER_X : Values of the counters X linked to the last job execution
- AMOUNT_X : Values of the amounts X linked to the last job execution

PCARD_TRACES Table

This table contains all errors or information messages generated during the job execution.

Content

Below themaintable columns:

- TASK_ID : batch identifier (*it is shown on thetrace file of thebatch*)
- PACKAGE_NAME and FUNCTION_NAME : PL-SQL Package and Function that generate the error
- USER_MESSAGE : Detail of the error
- ORA_ERROR_CODE : Oracle Error Code
- ERROR_CODE: Error code in PowerCARD format PWC-00000.

PCARD_MESSAGE_APP Table

It is the repository table of **PowerCARD** batches error codes batches PowerCARD .

Content

Below the main table columns:

- APP_CODE : Application Module
- ERROR_CODE : Error code in PowerCARD format PWC-00000
- ERROR_NAME : Name of the Error code
- MESSAGE : Detail of the error

Daemon logs and traces

Each Daemon generates its own Log File in \$TRACE Directory.

The trace file

This is the file having the name defined in the RESOURCE table. It is located in the \$TRACE directory.

Structure

Each log line follows the below logic:

- Date : Generation Date of the trace line(*DDMM Format*)
- Space ''
- Hour : Time of the trace line generation (*HH24MISSMMM format*)
- Space ''
- PPID : Unix Process Parent Id
- Space ''
- PID : Unix Process Id
- Space ''
- TID : Unix Thread Id
- Pipe '|'
- Trace Level : 0 (*HIGHEST*) to 5 (*LOWEST*)
- Pipe '|'
- Space ''
- Text: Content of the trace till the position 131, padded on the right with spaces.

Content

A daemon's log file is structured as follows:

- The header of the daemon startup

```
0208 132129979 00001 21233854 00001|5| *****
                                0208 132129979 00001 21233854 00001|5| **
    New Session Started          **
0208 132129979 00001 21233854 00001|5| *****
                                0208 132129979 00001 21233854 00001|5| Start
    ResetResource()
0208 132130212 00001 21233854 00001|5| Start   ResetResource()
0208 132130212 00001 21233854 00001|5| End ResetResource(OK)
0208 132130212 00001 21233854 00001|5| Start LogEvent(880000, INSR, )
0208 132130907 00001 21233854 00001|5| Start CommitOracle()
                                0208 132130907 00001 21233854 00001|5|
    End   CommitOracle(VOID)                                0208
132130907 00001 21233854 00001|5| End   LogEvent(OK)
0208 132130907 00001 21233854 00001|5| Start UpdateResource()
0208 132130907 00001 21233854 00001|5| Resource Id    = 880000
0208 132130907 00001 21233854 00001|5| Resource Live   = Y
0208 132130907 00001 21233854 00001|5| Processing Step = ONL
0208 132130922 00001 21233854 00001|5| End UpdateResource(OK)
0208 132130922 00001 21233854 00001|5| Start DisconnectOracle()
0208 132130925 00001 21233854 00001|5| End   DisconnectOracle(VOID)
0208 132130925 00001 21233854 00001|5| Start FreeOracleCtx()
                                0208 132130925 00001 21233854 00001|5| End
    FreeOracleCtx()
0208 132130926 00001 21233854 00515|5| Start ConnectOracle()
                                0208 132131257 00001 21233854 00515|5| End
    ConnectOracle(OK)
```

- A header/enqueue for every loop

```
0208      132131257 00001 21233854 00515|5| Start DaemonNotifProcess(16)
                                0208      132132004 00001 21233854
00515|5| End DaemonNotifProcess(OK)
```

The stat file

This file contains all display console or error of the shell program associated to the job, which would have occurred during the execution of the process. This log file is usually empty.

RESOURCES table

This is the table containing all the demons definitions, their parameters and characteristics.

Below the main table columns of this table:

- RESOURCE_ID : Daemon Identifier
- RESOURCE_CODE: Resource Code related to Daemon and used in the PL-SQL Programs related to the daemon.
- WORDING : Daemon wording or description
- OBJECT_NAME : Name of the executable file of the Daemon
- PRIS_OBJECT_PATH : Folder of the daemon executable file
- PRIS_START_SCRIPT_NAME : Name of the shell script used to start the Daemon
- PRIS_START_SCRIPT_PATH : Folder of the daemon shell start file
- PRIS_STOP_SCRIPT_NAME : Name of the shell script used to stop the Daemon
- PRIS_STOP_SCRIPT_PATH : Folder of the daemon shell stop file
- PRIS_BANK_CODE : The bank code for Daemons related to a specific Bank only
- PRIS_*_KEY_* : Name of the Keys used by the external cryptographic modules related to Daemons (*Example : HSM*)
- PRIS_LOG_FILE_NAME : Name of the log file of the Daemon
- PRIS_TIMER_ACQUIRER : Number of second of the sleep related to the Daemon
- PRIS_PID : Process ID (*PID*) of the Daemon in the Application Server

2.6.3 GUI (Graphical User Interface) logs and traces

The application traces of PowerCARD(*and stack traces*) are logged in the global JVM file of the WebSphere server "SystemOut.log".

The messages are following the (*was7*) format

```
[1/7/13 15:02:51:711 CST] 0000002f MessageBroker I
org.springframework.flex.servlet.MessageBrokerHandlerAdapter handle Channel
endpoint my-secure-amf received request.
```

Each line is interpreted as follows:

- timestamp

In the example, the timestamp is [1/7/13 15:02:51:711 CST].

The format of the timestamp is based on the location setup of the process server. It includes a complete date with millisecond precision and a time zone.

- Thread Identifier

In the example the thread ID is 0000002f. The thread ID is a hexadecimal code of 8 positions generated from the hash code of the thread that generates the message.

- Class

In the example the class is MessageBroker.

The class is the abbreviated name of the component that generates the message. For WAS, it will be the class of the component that generates the message, and for applications (*PWC*) the class of the object used to generate the message.

- Event Type

In the example the event type is I. The event type is a 1 character. Possible values are:

- F Fatal Error
- E Error
- W Warning
- A Audit message
- I Information message
- C Configuration message
- D Detail message
- O standard output console (*System.out*) of an application or an internal component of WAS
- R output error console (*System.err*) of an application or an internal component of WAS
- Z Unknown event type

- Message

In this example the message is

`org.springframework.flex.servlet.MessageBrokerHandlerAdapter` handle Channel endpoint `my-secure-amf` received request.

The message is the data that is traced to “SystemOut.log” by the component itself.

3 List of common batches

3.1 (transaction_ctrl_v3) Batch: Bank control

3.1.1 Batch flow

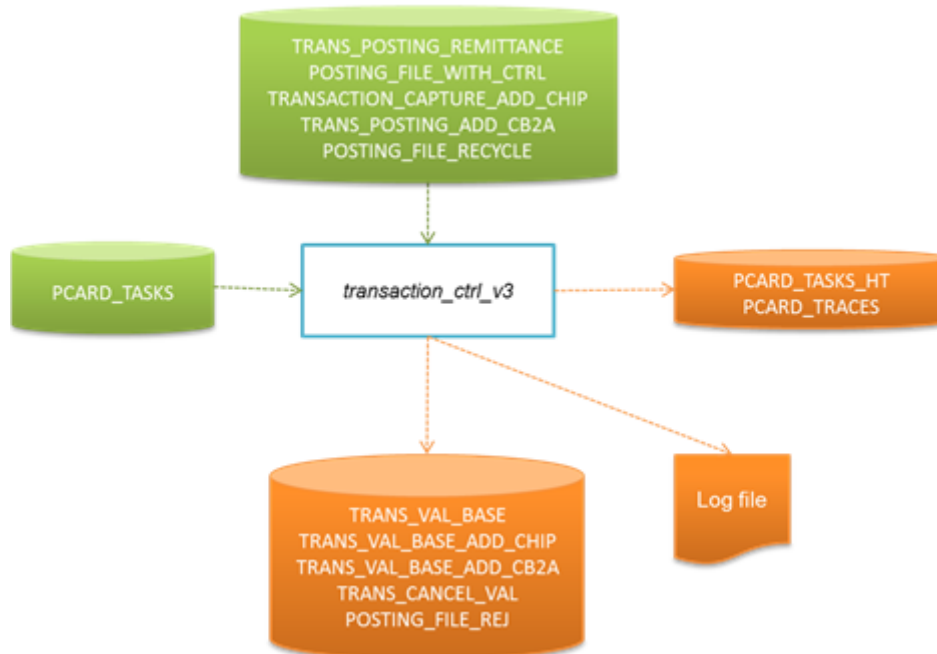


Figure 8: Banks controls batch

3.1.2 Batch description

This batch purpose is to check the transactions that have been loaded successfully during the posting file loading job against the list of functional control that are parametrized in PowerCARD. Each transaction that pass the controls is considered as valid transaction, the others are considered as rejected transactions and will be stored in the database while waiting for corrections.

3.1.3 Batch components

Shell	\$SHL/transaction_ctrl_v3.sh
Binary	\$BIN/transaction_ctrl_v3
File control	N/A
Task	transaction_ctrl_v3
Trace file	\$TRACE/transaction_ctrl_v3.trc000
Supports multi-cutoff	Yes, starting from PowerCARD V3.5
Command	transaction_ctrl_v3.sh <Trace file name><GROUP_CODE>

Default value when the variable <GROUP_CODE> is left blank will be "DFLGRP".

3.1.4 Batch parameters

Below are the fields parametrized in PCARD_TASKS table for this task:

Column name	Parameter	Description	Mandat.
BUSINESS_DATE	business_date	Force the processing date to this date [BUSINESS_DATE]	N

3.1.5 Parameters examples

Below are the fields parameterized in PCARD_TASKS table for this task:

Case	1
Description	Banks controls
Chain/recovery	Chain
Parent	load_posting_new_file
Child	pre_clearing

3.1.6 Input data

Type	Name	Description
Table	TRANS_POSTING_REMITTANCE	Remittance processed
Table	POSTING_FILE_WITH_CTRL	POS transactions loaded
Table	TRANSACTION_CAPTURE_ADD_CHIP	Transaction EMV data
Table	POSTING_FILE_RECYCLE	Recycled transactions
Table	TRANS_SPEC_LODGING	Lodging data

3.1.7 Output data

Type	Name	Description
Table	TRANS_VAL_BASE	Controlled transactions
Table	TRANS_VAL_BASE_ADD_CHIP	EMV data of controlled transactions
Table	TRANS_CANCEL_VAL	Cancelled transactions
Table	POSTING_FILE_REJ	Rejected POS transactions
Table	CRE_REM_CONTROL	CRE table containing the remittance controls results

3.1.8 Counters

The transaction control job stores the counters of the processed transactions in the log file which is created in \$TRACE directory of the application server. The log file will contain the following information:

- Number of processed transactions during the transaction control job
- Number of valid transactions
- Number of rejected transactions

```

----- [ TxLoaded   :%10d ]
----- [ TxRecycled :%10d ]
----- [ TxSuccess  :%10d ]
----- [ TxRejected :%10d ]
----- [ AmSuccess  :%10d ] --- Amount
----- [ AmRejected :%10d ]

```

AmSuccess : total amount of transactions successfully proceed

AmRejected : total amount of rejected transactions

In addition, there are CRE interface files that are generated in order to check the volume of data processed.

This CRE interface file is the result of the transaction control process on the different transaction batches. It contains:

- The transaction batch main characteristics, such as the date, currency, number of transactions and their type
- Merchant details
- Total number of valid transactions aggregated by transaction code and reversal indicator
- Total amount of valid transactions aggregated by transaction code and reversal indicator
- Total number of rejected transactions aggregated by transaction code and reversal indicator
- Total amount of rejected transactions aggregated by transaction code and reversal indicator

The flow of this CRE file is described in chapter 6 “(transaction_control_file) CRE generation of controlled remittance”

Reversal Summary

It is a CRE interface file that is generated for the reversals that have been processed during the transaction control job. It contains the detail of transaction cancellations captured by users and processed in PowerCARD. It enables the acquirer bank to trace the totals of reversals for each business day.

Aggregation is done by:

- Acquirer bank
- Transaction batch
- Transaction code

The flow of this CRE file is described in chapter 6 “(reversal_transaction_file) CRE reversal remittances generation batch”

Summary of Rejected Operations

This CRE interface file provides a reporting of the transaction rejections. It is the list of all transactions that are rejected by the transaction control job.

Aggregation is done by:

- Acquirer bank
- Transaction batch
- Transaction code

The flow of this CRE file is described in chapter 6 “(rejected_operation_file) CRE generation of rejected operations”

3.1.9 Warning

The batch may end with Warning in case some records have been rejected. Rejected records are loaded in PCARD_TRACES tables.

Return code listing: 00004, 00016, 00017, 00018, 00019, 00020, 00021, 00024, 00025, 00026, 00027, 00028, 00048, 00050, 00051, 00054, 00055, 00073, 01169, 01264, 01265, 15031, 15032

3.1.10 Recovery procedure

Re-run the bank controls batch.

Blocking	Yes
Clean	No
Mode	From the beginning
Batch	The same
Parameters	Case 01

3.2 (pre_clearing) Batch: Pre-clearing

3.2.1 Batch flow

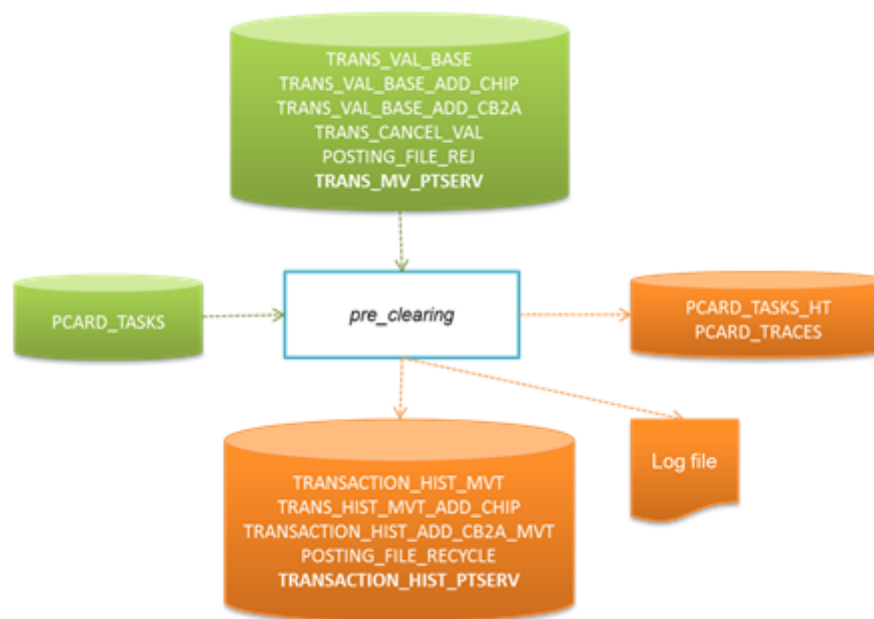


Figure 9: Pre-clearing batch

3.2.2 Batch description

Some of the processes done by this batch are:

- Preparing acquired transactions for clearing batches
- Computing interchange fees
- Creating transactions for card membership fees.

3.2.3 Batch components

Shell	\$SHL/pre_clearing.sh
Binary	\$BIN/pre_clearing
File control	N/A
Task	pre_clearing
Supports multi-cutoff	Yes, starting from PowerCARD V3.5
Trace file	\$TRACE/pre_clearing.trc000

Command	pre_clearing.sh <Trace file name><GROUP_CODE>
---------	---

Default value when the variable <GROUP_CODE> is left blank will be "DFLGRP".

3.2.4 Batch parameters

Below are the fields parametrized in PCARD_TASKS table for this task:

Column name	Parameter	Description	Mandat.
BUSINESS_DATE	business_date	Force the processing date to this date [BUSINESS_DATE]	N

3.2.5 Parameters examples

Below are the fields parameterized in PCARD_TASKS table for this task:

Case	1
Description	Pre clearing
Chain/recovery	Chain
Parent	transaction_ctrl
Child	visa_generate_outgoing ipm_generate_outgoing_file merchant_settlement_batch

3.2.6 Input

Type	Name	Description
Table	TRANS_VAL_BASE	Controlled transactions
Table	TRANS_VAL_BASE_ADD_CHIP	EMV data of controlled transactions
Table	TRANS_VAL_SPEC_LG	Valid lodging addendum data
Table	TRANS_CANCEL_VAL	Table of cancelled transactions
Table	POSTING_FILE_REJ	POS transactions controlled KO
Table	TRANS_PTSESV_MV	Point of sale transaction fees

3.2.7 Output data

Type	Name	Description
Table	TRANSACTION_HIST_MVT	Controlled transactions and pre-cleared
Table	TRANS_HIST_MVT_ADD_CHIP	EMV data of pre-cleared transactions
Table	POSTING_FILE_RECYCLE	Controlled transactions KO prepared for recycling
Table	CRE_REM_ACQUITTEMENT	CRE table of the remittance manually captured
Table	TRANS_HIST_MVT_LG	Lodging data

Type	Name	Description
Table	TRANSACTION_HIST_PTSEV	History of point of sale fee transactions

3.2.8 Counters

```

----- [ Processed :%10d]
----- [ Success :%10d]
----- [ Rejected :%10d]
----- [ AmProcessed :%10d]
----- [ AmSuccess :%10d]
----- [ AmRejected :%10d]

```

3.2.9 Warning

The batch may end with Warning in case some records have been rejected

A track of these records is kept in PCARD_TRACES table.

Return code listing: 00001, 00004, 00005, 00015, 00016, 00017, 00018, 00019, 00020, 00020, 00021, 00021, 00022, 00023, 00024, 00025, 00026, 00027, 00028, 00029, 00030, 00031, 00032, 00033, 00034, 00035, 00036, 00037, 00038, 00039, 00040, 00041, 00042, 00043, 00044, 00045, 00049, 00050, 00051, 00054, 00055, 00066, 00075, 01168, 01169, 01170, 01170, 01171, 01173, 01174, 01175, 01260, 01265, 01266, 01267, 01268, 99999.

3.2.10 Recovery procedure

Restart from the batch transaction_ctrl_v3

Blocking	Yes
Clean	No
Mode	From the beginning
Batch	transaction_ctrl_v3
Parameters	Case 01

3.3 (visa_generate_outgoing) Batch: generate Visa outgoing

3.3.1 Batch flow

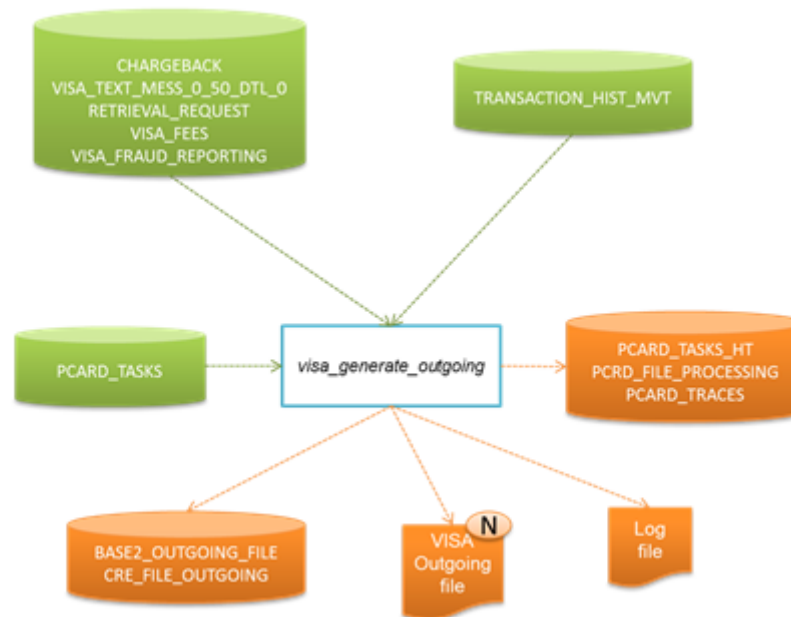


Figure 10: Visa outgoing batch

3.3.2 Batch description

The process generates the file to be submitted to Visa via BASEII. Its contents are:

- Presentment
- Chargeback
- Document request response
- Fraud
- Text message
- Retrieval requests
- Chargeback

The batch can be rerun as per VISA clearing entities defined in the system.

It can be run in parallel with other outgoing batches. A part of the process need to reserve a resource common for other outgoing batches and may need to wait that other batch unlock the resource.

The batch's parameters allow to select the format of the file generated, the file can be either EBDIC or in ASCII format.

The batch prepares also the data needed to generate the CRE outgoing files.

3.3.3 Batch components

Shell	\$SHL/visa_generate_outgoing.sh
Binary	\$BIN/visa_generate_outgoing
File control	N/A
Task	visa_generate_outgoing

Trace file	\$TRACE/visa_generate_outgoing.trc000
Supports multi-cutoff	Yes, starting from PowerCARD V3.5
Command	visa_generate_outgoing.sh <Trace file name><GROUP_CODE>

Default value when the variable <GROUP_CODE> is left blank will be "DFLGRP".

3.3.4 Batch parameters

Below are the fields parameterized in PCARD_TASKS table for this task:

Column name	Parameter	Description	Mandat
BUSINESS_DATE	business_date	Force the processing date to this date [BUSINESS_DATE]	N
PARAMETER_V1_DATA	Visa Data File Name	VISA outgoing files name which will have a suffix "clearing bin" added by the batch	Y
PARAMETER_V2_DATA	Separate Collection File	F (Financial ou C (Collection). Default value F	Y
PARAMETER_V3_DATA	Network Code	Network Code VISA (NETWORK.NETWORD_CODE) Default value 01	Y
PARAMETER_V4_DATA	Regulatory Context	Regulatory Context: 0 Base II privative 1 Base II banks in same groups 2 Base II banks in different groups 3 Base II under VISA network VISA). Default value is 0123, this parameter is used to control the behavior of the content of the base II file. Refer to the implementation specialist on this setting as it brings in concept of territoriality.	Y
PARAMETER_V5_DATA	Clearing Bin	Clearing Bin of the file to be generated	Y
PARAMETER_V6_DATA	File format Bin	File format of the outgoing file: ASCII for flat file in ASCII format EBDIC for EBDIC format	Y

3.3.5 Parameters examples

Below are the fields parametrized in PCARD_TASKS table for this task:

Case	1	
Description	Outgoing Visa	
Chain/recovery	Chain	
Parent	pre_clearing	

Child	Upload outgoing file to Visa edit package	
PARAM_V1	'VISA'	
PARAM_V2	'F'	
PARAM_V3	'01'	
PARAM_V4	'3'	
PARAM_V5	'466666'	

3.3.6 Input data

Type	Name	Description
Table	TRANSACTION_HIST_MVT	Controlled transaction and pre-cleared 1st and 2nd presentment to the network
Table	CHARGEBACK	1st and 2nd chargeback
Table	VISA_TEXT_MESS_0_50_DTL_0	Network text message
Table	RETRIEVAL_REQUEST	Documents requests response
Table	VISA_FEES	Visa fees

3.3.7 Output data

Type	Name	Description
File	« PARAMETER_V1_DATA »<CLEARING_BIN>	Clearing BIN of the Base II outgoing entity for which the file is be generated
Table	CRE_FILE_OUTGOING	Outgoing CRE table
Table	CRE_OP_OUTGOING_CYCLE	Operations cycle CRE table

Name	Type	Size	Format
« PARAMETER_V1_DATA »<CLEARING_BIN>	Record, Fixe, ebcdic	168	Base II

No file is generated if there is no entry to generate.

3.3.8 Counters

```

----- [ Processed :%d ]
----- [ Success :%d ]
----- [ Rejected :%d ]
----- [ Am Processed :%d ]
----- [ Am Succeeded :%d ]
----- [ Am Rejected :%d ]

```

3.3.9 Warning

The batch ends either OK or NOK.

Return code listing: 00004, 00016, 00017, 00018, 00019, 00021, 00025, 00026, 00027, 00028, 00041, 00054, 00055, 01169, 01175, 03103, 04526, 99998, 99999.

3.3.10 Recovery procedure

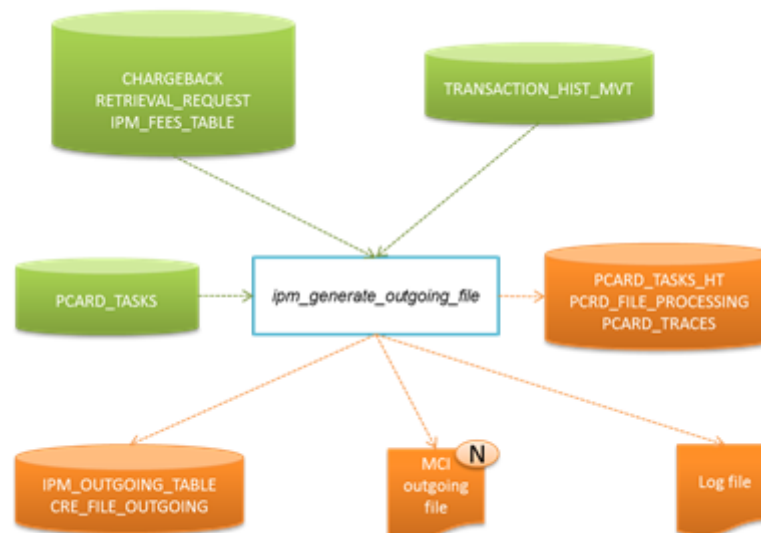
If the batch is rerun, files which were not generated in the previous batch will be generated.

Blocking	Yes
Clean	No
Mode	Where it has been stopped
Batch	The same
Parameters	Case 01

3.4 (ipm_generate_outgoing_file) Batch: MasterCard outgoing

3.4.1 Batch flow

Figure 11: MCI outgoing batch



3.4.2 Batch description

The process generates records in IPM format:

- Presentment
- Chargeback
- Document request responses
- 2nd presentment
- Retrieval requests

The batch can be rerun as many as MasterCard clearing entities defined in the system.

It can be run in parallel with other outgoing batches. A part of the process need to reserve a resource common to other outgoing batches and may need to wait that other batch unlock the resource.

3.4.3 Batch components

Shell	\$SHL/ipm_generate_outgoing_file.sh
Binary	\$BIN/ipm_generate_outgoing_file
File control	N/A
Task	ipm_generate_outgoing_file
Trace file	\$TRACE/ipm_generate_outgoing_file.trc000
Supports multi-cutoff	Yes, starting from PowerCARD V3.5
Command	ipm_generate_outgoing_file.sh <Trace file name><GROUP_CODE>

Default value when the variable <GROUP_CODE> is left blank will be "DFLGRP".

3.4.4 Batch parameters

Below are the fields parametrized in PCARD_TASKS table for this task:

Column name	Parameter	Description	Mandat
BUSINESS_DATE	business_date	Force the processing date to this date [BUSINESS_DATE]	N
PARAMETER_V1_DATA	Generate GCMS file	IPM file format : 'G_ASCII_1014' : If you want to generate the file in ASCII format 1014 'G_EBCDI_1014': If you want to generate the file in EBCDIC format 1014 default value « G_EBCDI_1014 »	Y
PARAMETER_V2_DATA	GCMS file name	Name of output IPM	Y
PARAMETER_V3_DATA	Collection/Financial	F (Financial) or C (Collection). Default value F	Y
PARAMETER_V4_DATA	Collection file name	Collection file name output of IPM batch	Y
PARAMETER_V5_DATA	Network Code	MasterCard Network code (NETWORK.NETWORK_CODE) Default value "02"	Y
PARAMETER_V6_DATA	Regulatory Context	Regulatory context: (0 IPM privative, 3 IPM MasterCard network). Default value3	Y

3.4.5 Parameters examples

Below are the fields parametrized in PCARD_TASKS table for this task:

Case	1	
Description	Outgoing MCI	
Chain/recovery	Chain	
Parent	pre_clearing	

Child	Submit ipm outgoing file to MIP	
PARAM_V1	'G_EBCDI_1014'	
PARAM_V2	'MCI_OUT'	
PARAM_V3	'F'	
PARAM_V4	NULL	
PARAM_V5	'02'	
PARAM_V6	'3'	

3.4.6 Input data

Type	Name	Description
Table	TRANSACTION_HIST_MVT	Transactions controlled and ready for clearing as 1st or 2nd presentment depending on the network.
Table	CHARGEBACK	1st and 2nd unpaid transactions
Table	RETRIEVAL_REQUEST	Documents requests responses
Table	IPM_FEES_TABLE	MasterCard fees

3.4.7 Output data

Type	Name	Description
File	« PARAMETER_V2_DATA »	Outgoing IPM file
Table	CRE_FILE_OUTGOING	Table CRE des outgoings
Table	CRE_OP_OUTGOING_CYCLE	CRE Table of operations cycle

Name	Type	Size	Format
« PARAMETER_V2_DATA »	Record, variable bloc	1014	IPM

No file is generated if there is no entry to generate.

3.4.8 Counters

```

----- [ Processed :%d ]
----- [ Success :%d ]
----- [ Rejected :%d ]

```

3.4.9 Warning

The batch ends either OK or NOK.

Return code listing:00004, 00015, 00016, 00016, 00017, 00018, 00019, 00021, 00025, 00026, 00027, 00028, 00054, 00055, 00074, 01169, 01175, 03103, 03105, 99999.

3.4.10 Recovery procedure

Blocking	Yes
Clean	No

Mode	From the beginning
Batch	The same
Parameters	Case 01

3.5 (safe_generate_fdx) Batch: Generation SAFE file FDX

3.5.1 Batch flow

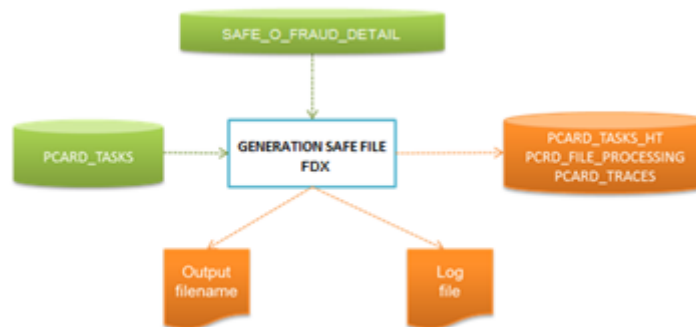


Figure 12: SAFE_GENERATE_FDX batch

3.5.2 Batch description

This batch generates the file containing fraudulent MasterCard transactions must be reported to SAFE.

- Fraud Add (FDA);
- Fraud Change (FDC);
- Fraud Delete (FDD);
- Fraud Confirmation (FDE).

3.5.3 Batch components

Shell	\$SHL/safe_generate_fdx.sh
Binary	\$BIN/ safe_generate_fdx
Control file	NA
Task	safe_generate_fdx
Trace file	\$TRACE/ SAFE_OUT_FDX.TRC000
Pre Command	Update pcard_tasks Set BUSINESS_DATE = (select LAST_CUTOFF_DATE_BO from switch_cut_off), START_DATE = sysdate , status = 'P' , where task_name = ' safe_generate_fdx ' and bank_code =\$Bank_code; Commit; Note this script can be replaced by SHL
Command	safe_generate_fdx.sh <Trace file name>

3.5.4 Batch parameters

Below are the fields parametrized in PCARD_TASKS table of this task:

Column name	Parameter	Description	Mandat.
PARAM_V1	Filename	Output filename	Y

3.5.5 Parameter examples

Below are the fields parametrized in PCARD_TASKS table of this task:

Case	1	
Description	This batch generates the file containing fraudulent MasterCard transactions must be reported to SAFE	
Chain/recovery		
Parent		
PARAM_V1	SAFE_OUT	
PARAM_V2	C	

3.5.6 Input data

Type	Name	Description
Table	SAFE_O_FRAUD_DETAIL	

3.5.7 Output data

Type	Name	Description
File	SAFE_OUT	

3.5.8 Counters

```
----- [ Processed :%d ]
```

The numbers of lines processed are displayed. Header and trailer excluded.

3.5.9 Warning

The process ends either OK or NOK.

Return code listing: 4501, 4519, 4521, 4524, 4526, 4527, 4529, 4590, 4999, 5608

Batch status query:

```
"SELECT status, ending_status, user_message from pcard_tasks where task_name = 'safe_generate_fdx' and bank_code = $Bank_code;"
```

The trace file query is as follows:

```
"SELECT trace_file_name from pcard_tasks where task_name = 'safe_generate_fdx' "
```

3.5.10 Recovery procedure

Need to re-run the batch. Transactions that were not processed in the previous batch are included in the current batch.

Blocking	Yes
Clean	No
Mode	Crash
Batch	The same
Parameters	Case 01

3.6 (safe_generate_fdn) Batch: Generation SAFE file FDN

3.6.1 Batch flow

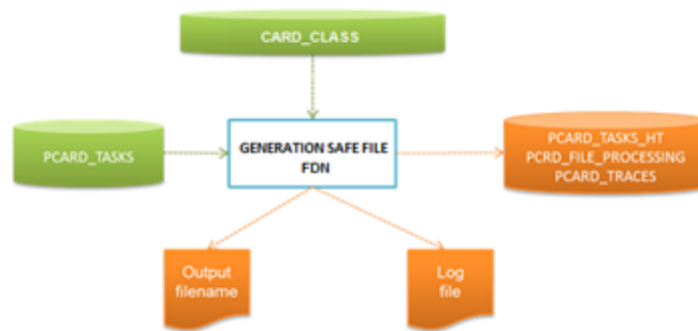


Figure 13: SAFE_GENERATE_FDN batch

3.6.2 Batch description

If there is no fraud to report for a specific calendar month, issuers must submit a Fraud Negative Report (FDN) record by the end of the reporting month. This batch generates the file containing FDN.

3.6.3 Batch components

Shell	\$SHL/safe_generate_fdn.sh
Binary	\$BIN/ safe_generate_fdn
Control file	NA
Task	safe_generate_fdn
Trace file	\$TRACE/ SAFE_OUT_FDN.TRC000
Pre Command	Update pcard_tasks Set BUSINESS_DATE = (select LAST_CUTOFF_DATE_BO from switch_cut_off), START_DATE = sysdate , status = 'P' , where task_name = ' safe_generate_fdn ' and bank_code =\$Bank_code; Commit;

	Note this script can be replaced by SHL
Command	safe_generate_fdn.sh <Trace file name>

3.6.4 Batch parameters

Below are the fields parametrized in PCARD_TASKS table of this task:

Column name	Parameter	Description	Mandat.
PARAM_V1	Filename	Output filename	Y

3.6.5 Parameters examples

Below are the fields parametrized in PCARD_TASKS table of this task:

Case	1	
Description	An FDN record should be submitted during the month for which there is no fraud to report.	
Chain/recovery		
Parent		
PARAM_V1	SAFE_OUT	

3.6.6 Input data

Type	Name	Description
Table	CARD_CLASS	

3.6.7 Output data

Type	Name	Description
File	Output file	

3.6.8 Counters

```
----- [ Processed :%d ]
```

The numbers of lines processed are displayed. Header and trailer excluded.

3.6.9 Warning

The process ends either OK or NOK.

Return code listing: 4501, 4519, 4521, 4524, 4526, 4527, 4529, 4590, 4999, 5608

Batch status query:

```
"SELECT status, ending_status, user_message from pcard_tasks where task_name = 'safe_generate_fdn' and bank_code = $Bank_code;"
```

The trace file query is as follows:

```
"SELECT trace_file_name from pcard_tasks where task_name = 'safe_generate_fdn' "
```

3.6.10 Recovery procedure

Need to re-run the batch. Transactions that were not processed in the previous batch are included in the current batch.

Blocking	Yes
Clean	No
Mode	Crash
Batch	The same
Parameters	Case 01

3.7 (archive_transaction) Batch: Archiving batch

3.7.1 Batch flow

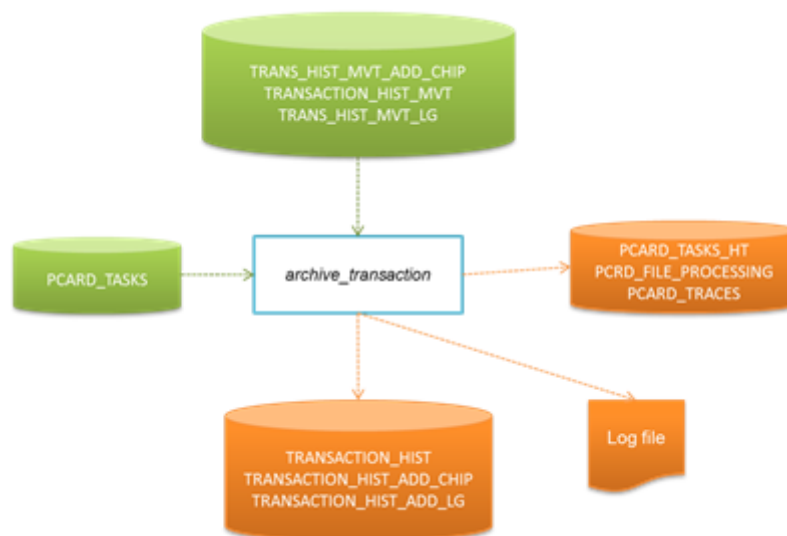


Figure 14: Transaction archiving batch

3.7.2 Batch description

The batch archives the acquired transactions that have been correctly controlled and sent for clearing.

3.7.3 Batch components

Shell	\$SHL/archive_transaction.sh
Binary	\$BIN/archive_transaction
Control file	N/A
Task	archive_transaction
Trace file	\$TRACE/archive_transaction.trc000
Command	archive_transaction.sh <Trace file name>

3.7.4 Batch parameters

Below are the fields parametrized in PCARD_TASKS table for this task:

Column name	Parameter	Description	Mandat.
BUSINESS_DATE	business_date	Force the processing date to this date [BUSINESS_DATE]	N
PARAMETER_V1_DATA	Recovery Mode	'N' Normal (default), 'R' Recovery	N
PARAMETER_V2_DATA			
PARAMETER_V3_DATA			

3.7.5 Parameters examples

Below are the fields parametrized in PCARD_TASKS table for this task:

Case	1	2
Description	Archiving transactions	Archiving recovery
Chain/recovery	Chain	Out of chain
Parent		'R'
PARAM_V1		
PARAM_V2		
PARAM_V3		

3.7.6 Input data

Type	Name	Description
Table	TRANSACTION_HIST_MVT	Transactions controlled and cleared as 1st or 2nd presentment depending on the network.
Table	TRANS_HIST_MVT_ADD_CHIP	EMV data of transactions in MVT history table.
Table	TRANS_HIST_MVT_LG	Lodging data of transactions in MVT history table

3.7.7 Output data

Type	Name	Description
Table	TRANSACTION_HIST	Transactions history
Table	TRANSACTION_HIST_ADD_CHIP	Transactions history of CHIP data
Table	TRANSACTION_HIST_ADD_LG	Transactions history of Lodging data

3.7.8 Counters

```

----- [ Processed :%d ]
----- [ Success :%d ]
----- [ Rejected :%d ]

```

3.7.9 Warning

The batch ends either OK or NOK.

Return code listing: 00004, 00017, 00019, 00028

3.7.10 Recovery procedure

Need to re-run the batch

Blocking	No
Clean	No
Mode	From the beginning
Batch	The same
Parameters	Case 02

3.8 (authorization_batch) Batch: Authorizations archiving

3.8.1 Batch flow

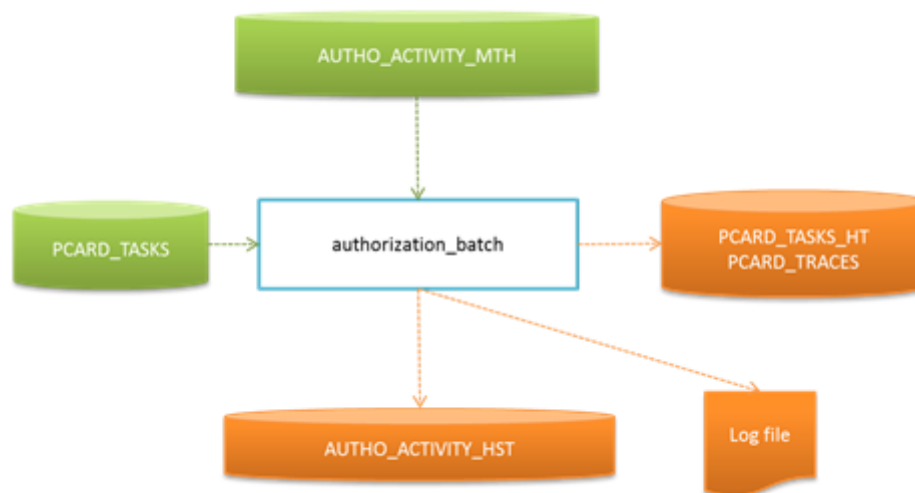


Figure 15: Authorization archiving batch

3.8.2 Batch description

This batch archives the authorizations matched with transactions, and the one that were not matched and exceed a certain number of days.

3.8.3 Batch components

Shell	\$SHL/authorization_batch.sh
Binary	\$BIN/authorization_batch
Control file	N/A
Task	authorization_batch
Trace file	\$TRACE/authorization_batch.trc000
Supports multi-cutoff	Yes, starting from PowerCARD V3.5
Command	authorization_batch.sh <Trace file name><GROUP_CODE>

Default value when the variable <GROUP_CODE> is left blank will be "DFLGRP".

3.8.4 Batch parameters

Below are the fields parametrized in PCARD_TASKS table for this task:

Column name	Parameter	Description	Mandat.
BUSINESS_DATE	business_date	Force the processing date to this date [BUSINESS_DATE]	N
PARAMETER_V1_DATA	Minutes to check in	1 default	Y
PARAMETER_V2_DATA	Number before commit	Number of authorizations records to archive before a commit	Y

3.8.5 Parameters examples

Below are the fields parametrized in PCARD_TASKS table for this task:

Case	1	
Description	Archiving authorizations	
Chain/recovery	Chain	
Parent		
PARAM_V1	1	
PARAM_V2	20000	

3.8.6 Input data

Type	Name	Description
Table	AUTHO_ACTIVITY_MTH	Table of authorizations

3.8.7 Output data

Type	Name	Description
Table	AUTHO_ACTIVITY_HST	Archived authorizations

3.8.8 Counters

```

----- [ Processed :%d ]
----- [ Success :%d ]
----- [ Rejected :%d ]

```

3.8.9 Warning

The batch ends either OK or NOK.

Return code listing: 00001, 00004, 00016, 00017, 00018, 00019, 00020, 00021, 00022, 00024, 00025, 00026, 00027, 00028, 00050, 00051, 00054, 00055, 01174, 03270, 99999

3.8.10 Recovery procedure

Need to re-run the batch.

Blocking	No
----------	----

Clean	No
Mode	From the beginning
Batch	Same
Parameters	

3.9 (visa_load_incoming) Batch: Load VISA incoming

3.9.1 Batch flow

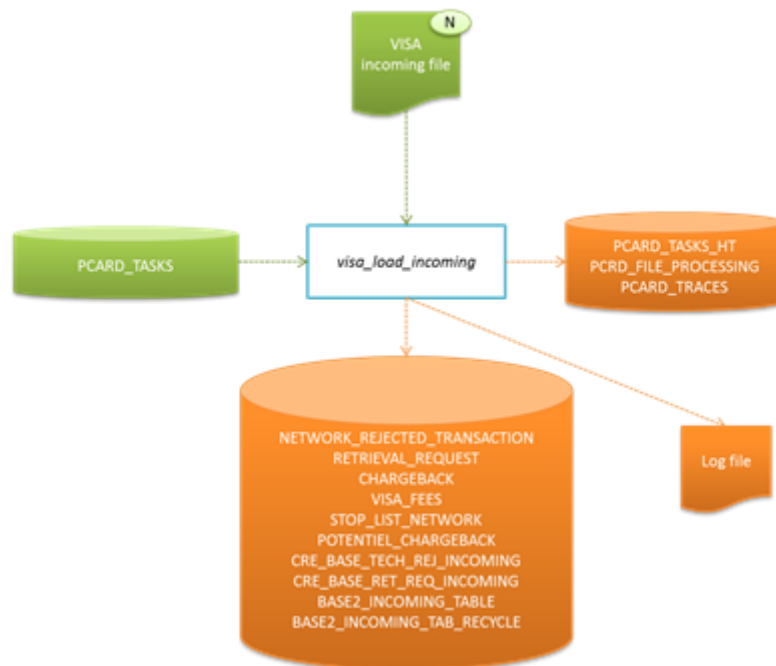


Figure 16: Load Visa incoming batch

3.9.2 Batch description

The batch aims to load transaction and chargeback records, fees, document request, VISA rejected records in the database to process them.

The batch can be run in parallel with the other incoming batches but it cannot be run twice at the same time

3.9.3 Batch components

Shell	\$SHL/visa_load_incoming.sh
Binary	\$BIN/visa_load_incoming
File control	N/A
Task	visa_load_incoming
Trace file	\$TRACE/visa_load_incoming.trc000
Supports multi-cutoff	Yes, starting from PowerCARD V3.5
Command	visa_load_incoming.sh <Trace file name> <input filename><GROUP_CODE>

Default value when the variable <GROUP_CODE> is left blank will be "DFLGRP".

3.9.4 Batch parameters

Below are the fields parametrized in PCARD_TASKS table for this task:

Column name	Parameter	Description	Mandat.
BUSINESS_DATE	business_date	Force the processing date to this date [BUSINESS_DATE]	N
PARAMETER_V1_DATA	Data file name	VISA incoming file name to be load in the system.	Y
PARAMETER_V2_DATA	File Record length	Record length. Default value 169	Y
PARAMETER_V3_DATA	Processing mode	Processing mode of the incoming files ('N' Normal, 'R' Recycling). N default value.	Y

3.9.5 Parameters examples

Below are the fields parametrized in PCARD_TASKS table for this task:

Case	1	2
Description	incoming VISA	Recycling VISA incoming file
Chain/recovery	Chain	Recovery
Parent		
PARAM_V1	'VISA_IN'	'VISA_IN'
PARAM_V2	'169'	'169'
PARAM_V3	'N'	'R'

3.9.6 Input data

Type	Name	Description
File	« PARAMETER_V1_DATA »	VISA incoming file

Name	Type	Size	Format
« PARAMETER_V1_DATA »*	Record, ebcdic, fixe	168	Base II

3.9.7 Output data

Type	Name	Description
Table	NETWORK_REJECTED_TRANSACTION	VISA rejected records
Table	CHARGEBACK	Chargebacks
Table	RETRIEVAL_REQUEST	Retrieval requests
Table	VISA_FEES	VISA fees
Table	BASE2_INCOMING_TAB_RECYCLE	Incoming transactions that failed the batch controls

3.9.8 Counters

```
----- [ Processed :%d ]
----- [ Success :%d ]
----- [ Rejected :%d ]
```

3.9.9 Counters

```
----- [ Processed :%d ]
----- [ Success :%d ]
----- [ Rejected :%d ]
```

3.9.10 Recovery procedure

Recycling the incoming and rerun the batch with « Processing mode » set to '**R**'.

Blocking	No
Clean	No
Mode	From stop point
Batch	The same
Parameters	Case 02

3.10 (processing_ipm_incoming_file) Batch: MasterCard incoming file load

3.10.1 Batch flow

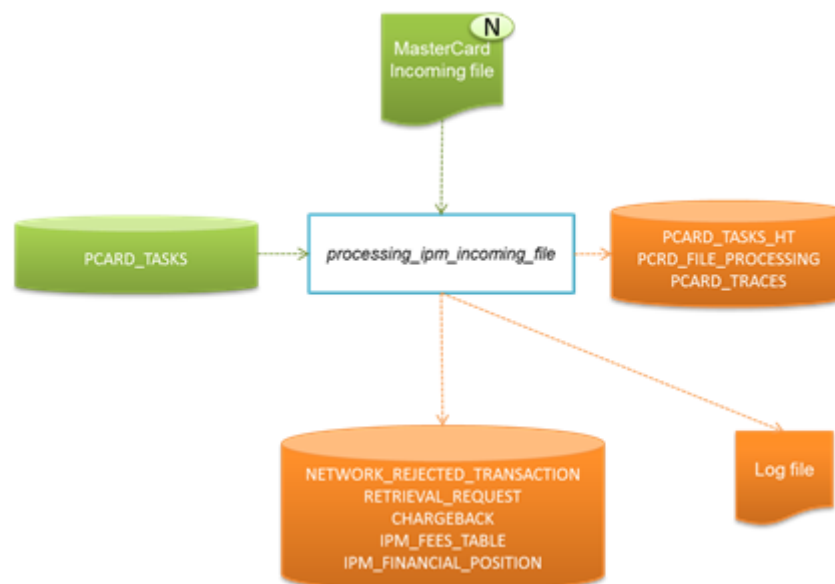


Figure 17: Load MasterCard incoming batch

3.10.2 Batch description

The batch loads transaction and chargebacks records, retrieval requests and MasterCard rejected records in the system to be processed.

The batch can be run in parallel with the other incoming batches but it cannot be run twice at the same time.

3.10.3 Batch components

Shell	\$SHL/processing_ipm_incoming_file.sh
Binary	\$BIN/processing_ipm_incoming_file
Control file	N/A
Task	processing_ipm_incoming_file
Trace file	\$TRACE/processing_ipm_incoming_file.trc000
Supports multi-cutoff	Yes, starting from PowerCARD V3.5
Command	processing_ipm_incoming_file.sh <Trace file name> <input filename><GROUP_CODE>

Default value when the variable <GROUP_CODE> is left blank will be "DFLGRP".

3.10.4 Batch parameters

Below are the fields parametrized in PCARD_TASKS table for this task:

Column name	Parameter	Description	Mandat.
BUSINESS_DATE	business_date	Force the processing date to this date [BUSINESS_DATE]	N
PARAMETER_V1_DATA	FILE NAME	Refers to the MasterCard incoming file name	Y
PARAMETER_V3_DATA	LOAD_CROSS_RATE	Load MasterCard conversion rates. Default value Y.	Y
PARAMETER_V4_DATA	USE_CROSS_RATE	Use of MasterCard conversion rates. Default value Y.	Y
PARAMETER_V5_DATA	FORMAT	Incoming file format. Possible values • ASCII_1014 • EBCDIC1014 Default value: ASCII_1014.	Y
PARAMETER_V6_DATA	PROCESSING_MODE	Processing mode of the incoming files ('N' Normal, 'R' Recycling). N default value.	Y

3.10.5 Parameters examples

Below are the fields parametrized in PCARD_TASKS table for this task:

Case	1	2
Description	incoming MCD	Recovery incoming MCD
Chain/recovery	Chain	Recovery

Parent		
PARAM_V1	'IPM_IN'	'IPM_IN'
PARAM_V2	NULL	NULL
PARAM_V3	'Y'	'Y'
PARAM_V4	'Y'	'Y'
PARAM_V5	'ASCII_1014'	'ASCII_1014'
PARAM_V6	'N'	'R'

3.10.6 Input data

Type	Name	Description
File	« PARAMETER_V1_DATA »	File MasterCard incoming

Name	Type	Size	Format
« PARAMETER_V1_DATA »*	Record, variable	1014	IPM

3.10.7 Output data

Type	Name	Description
Table	NETWORK_REJECTED_TRANSACTION	MasterCard rejected records
Table	CHARGEBACK	Chargeback
Table	RETRIEVAL_REQUEST	Retrieval request
Table	IPM_FEES_TABLE	MasterCard fees
Table	IPM_FINANCIAL_POSITION	MasterCard Financial activity summary of the member
Table	IPM_INCOMING_RECYCLE	Incoming transaction that have failed the controls

3.10.8 Counters

```

----- [ Processed :%d ]
----- [ Success :%d ]
----- [ Rejected :%d ]

```

3.10.9 Warning

The batch can end with Warning, if some records fail the batch controls. These records are stored in IPM_INCOMING_RECYCLE table.

Return code listing: 00001, 00004, 00015, 00016, 00017, 00018, 00019, 00020, 00021, 00025, 00026, 00027, 00028, 00030, 00043, 00049, 00050, 00051, 00054, 00055, 00075, 01169, 04526, 04590, 04597, 04598, 04699, 15003, 15006, 15008, 15031, 15032, 99999.

3.10.10 Recovery procedure

Recycling the incoming and rerun the batch with « Processing mode » set to '**R**'

Blocking	No
Clean	No

Mode	From stop point
Batch	The same
Parameters	Case 02

3.11 (adm_update_business_date_bo) Batch: update business date

3.11.1 Batch flow

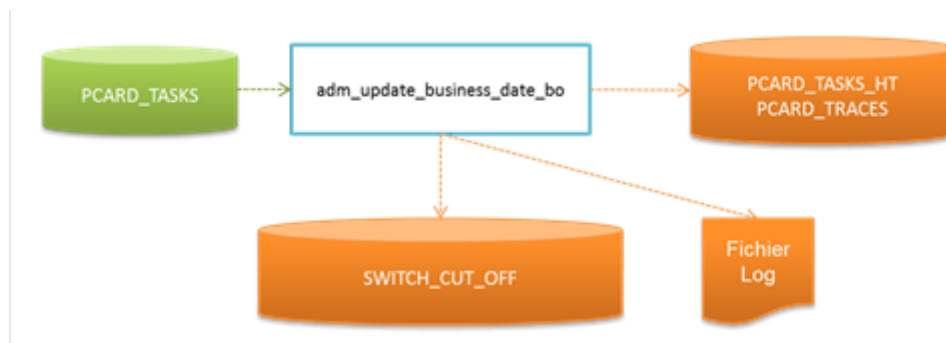


Figure 18: Update business date batch

3.11.2 Batch description

The batch is used to increment the **PowerCARD** business_date + one day.

3.11.3 Batch components

Shell	\$SHL/adm_update_business_date_bo.sh
Binary	\$BIN/adm_update_business_date_bo
Control file	N/A
Task	adm_update_business_date_bo
Trace file	\$TRACE/adm_update_business_date_bo.trc
Supports multi-cutoff	Yes, starting from PowerCARD V3.5
Command	adm_update_business_date_bo <Trace file name><GROUP_CODE>

Default value when the variable <GROUP_CODE> is left blank will be "DFLGRP".

3.11.4 Batch parameters

Below are the fields parametrized in PCARD_TASKS table for this task:

Column name	Parameter	Description	Mandat.
BUSINESS_DATE	business_date	Force the processing date to this date [BUSINESS_DATE]	N

3.11.5 Parameters examples

Below are the fields parametrized in PCARD_TASKS table for this task:

Case	1	
------	---	--

Description	Conversion	
Chain/recovery	Chain	

3.11.6 Input data

Not applicable.

3.11.7 Output data

Type	Name	Description
Table	SWITCH_CUT_OFF	Table where backoffice business_date of the system is stored

3.11.8 Counters

There are no counters assigned to this process.

3.11.9 Warning

The batch ends either OK or NOK.

Return code listing: 00004, 00017, 00021, 00042

3.11.10 Recovery procedure

Need to rerun the batch

Blocking	Yes
Clean	No
Mode	From the beginning
Batch	The same
Parameters	Case 01

3.12 (generate_statistics) Batch: Generate statistics

3.12.1 Batch flow

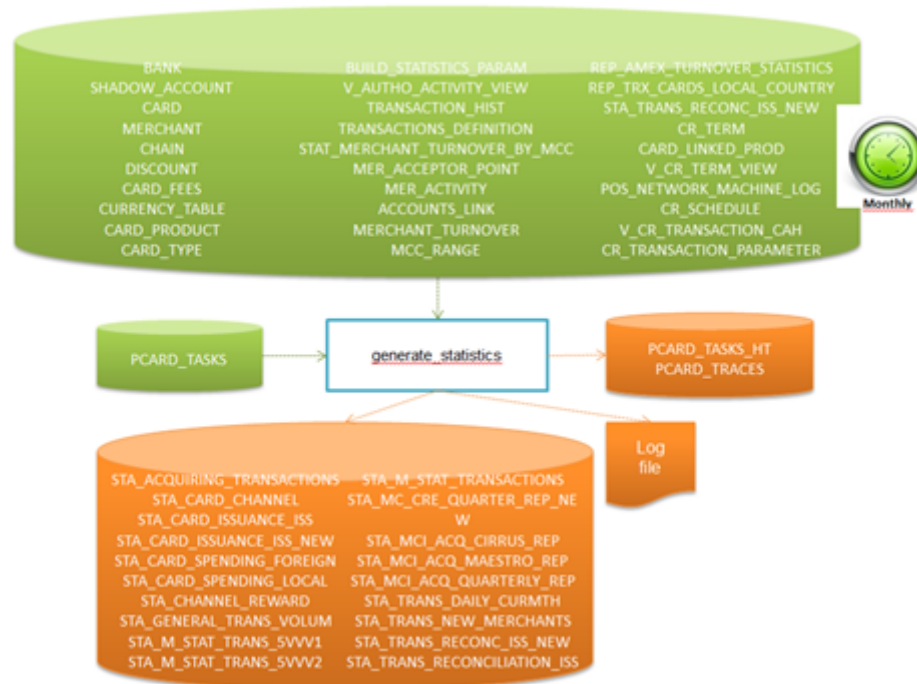


Figure 19: Generate statistics batch

3.12.2 Batch description

This batch is used to collect various statistics on the processing performed by the Platform. It is a monthly batch that collects acquiring and issuing indicators used in other processes. It is notably a requirement for the quarterly reports.

3.12.3 Batch components

Shell	\$SHL/generate_statistics.sh
Binary	N/A
Control file	N/A
Task	generate_statistics
Trace file	\$TRACE/generate_statistics.trc000
Command	generate_statistics.sh <Trace file name>

3.12.4 Batch parameters

Below are the fields parametrized in PCARD_TASKS table of this task:

Column name	Parameter	Description	Mandat.
BUSINESS_DATE	business_date	Force the processing date to this date [BUSINESS_DATE]	N
PARAM_V1	Period	Month in format 'MMYYYY'	M

3.12.5 Parameters examples

Below are the fields parametrized in PCARD_TASKS table of this task.

Case	1	
Description	generate_statistics	
Chain/recovery	Chain	
Parent	End of monthly batches	
Child	Report generation	
PARAM_V1	042014	

Notes on the statistic batch:

- The batch computes monthly indicators based on several criteria:
 - Authorization
 - Chargebacks
 - Transactions (Issuing / Acquiring)
- The batch computes the indicators for all the Institution defined in the platform
- The batch can be set to execute on past months without issues, it will erase and replace the corresponding data and period
- The currency needs to be set per Institution
 - The currency needs to be set with HPS during setup
 - Currency may be changed, it will require to re execute the batch for each month

3.12.6 Input data

Type	Name	Description
Table	BANK	
Table	SHADOW_ACCOUNT	
Table	CARD	
Table	MERCHANT	
Table	CHAIN	
Table	DISCOUNT	
Table	CARD_FEES	
Table	CURRENCY_TABLE	
Table	CARD_PRODUCT	
Table	CARD_TYPE	
Table	BUILD_STATISTICS_PARAM	
Table	V_AUTHO_ACTIVITY_VIEW	
Table	TRANSACTION_HIST	
Table	TRANSACTIONS_DEFINITION	
Table	STAT_MERCHANT_TURNOVER_BY_MCC	
Table	MER_ACCEPTOR_POINT	
Table	MER_ACTIVITY	

Type	Name	Description
Table	ACCOUNTS_LINK	
Table	MERCHANT_TURNOVER	
Table	MCC_RANGE	
Table	REP_AMEX_TURNOVER_STATISTICS	
Table	REP_TRX_CARDS_LOCAL_COUNTRY	
Table	STA_TRANS_RECONC_ISS_NEW	
Table	CR_TERM	
Table	CARD_LINKED_PROD	
Table	V_CR_TERM_VIEW	
Table	POS_NETWORK_MACHINE_LOG	
Table	CR_SCHEDULE	
Table	V_CR_TRANSACTION_CAH	
Table	CR_TRANSACTION_PARAMETER	

3.12.7 Output data

Type	Name	Description
Table	STA_ACQUIRING_TRANSACTIONS	
Table	STA_CARD_CHANNEL	
Table	STA_CARD_ISSUANCE_ISS	
Table	STA_CARD_ISSUANCE_ISS_NEW	
Table	STA_CARD_SPENDING_FOREIGN	
Table	STA_CARD_SPENDING_LOCAL	
Table	STA_CHANNEL_REWARD	
Table	STA_GENERAL_TRANS_VOLUM	
Table	STA_M_STAT_TRANS_5VVV1	
Table	STA_M_STAT_TRANS_5VVV2	
Table	STA_M_STAT_TRANSACTIONS	
Table	STA_MC_CRE_QUARTER_REP_NEW	
Table	STA_MCI_ACQ_CIRRUS_REP	
Table	STA_MCI_ACQ_MAESTRO_REP	
Table	STA_MCI_ACQ_QUARTERLY_REP	
Table	STA_TRANS_DAILY_CURMTH	
Table	STA_TRANS_NEW_MERCHANTS	
Table	STA_TRANS_RECONC_ISS_NEW	
Table	STA_TRANS_RECONCILIATION_ISS	
Table	STA_TRANS_VOLUM_BYMCCGROUP	
Table	STA_TRANS_VOLUM_CHAIN	

Type	Name	Description
Table	STA_TRANS_VOLUM_CHAINMCCGP	
Table	STA_TRANS_VOLUM_CHAINMERDTL	
Table	STA_TRANS_VOLUM_MEROUTDTL	
Table	STA_TRANS_VOLUM_TOP10OUT_INT	
Table	STA_TRANS_VOLUM_TOP10OUTLET	

3.12.8 Counters

```
----- [ Processed :%d ]
```

The numbers of lines processed are displayed. Header and trailer excluded.

3.12.9 Warning

The batch ends either OK or NOK.

3.12.10 Warning

The batch ends either OK or NOK.

3.13 (accounting_batch) Batch: Accounting batch

3.13.1 Batch flow

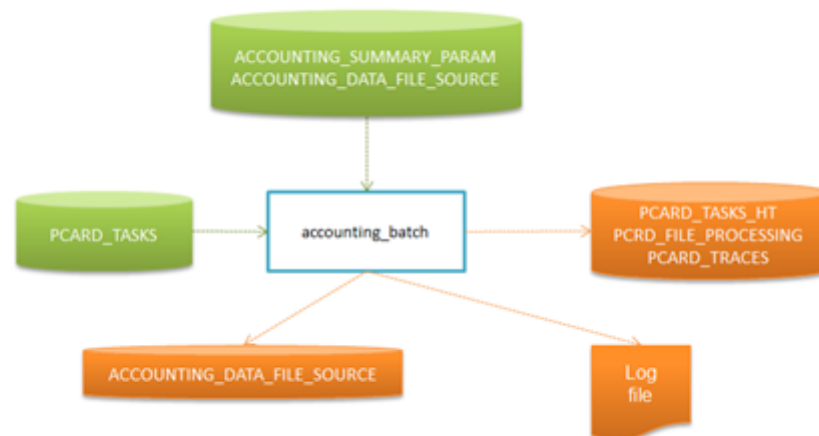


Figure 20: Accounting_batch

3.13.2 Batch description

This batch generates the accounting entries defined within **PowerCARD**. It is used to feed the next batch `gen_accounting_file`.

3.13.3 Batch components

Shell	\$SHL/ accounting_batch.sh
Binary	\$BIN/ accounting_batch.bin
Control file	NA

Task	accounting_batch
Trace file	\$TRACE/ accounting_batch.trc000
Pre Command	Update pcard_tasks Set BUSINESS_DATE = (select LAST_CUTOFF_DATE_BO from switch_cut_off), START_DATE = sysdate , status = 'P' , where task_name = 'accounting_batch' and bank_code = \$Bank_code; Commit; Note: this script can be replaced by shl
Supports multi-cutoff	Yes, starting from PowerCARD V3.5
Command	accounting_batch.sh <Trace file name><GROUP_CODE>

Default value when the variable <GROUP_CODE> is left blank will be "DFLGRP".

3.13.4 Batch parameters

Below are the fields parametrized in PCARD_TASKS table of this task:

Column name	Parameter	Description	Mandat.
PARAM_V1	Filename	Input filename : depreciated not used	N
PARAM_V2	Bank Code	Bank code	Y

3.13.5 Parameter examples

Below are the fields parametrized in PCARD_TASKS table of this task.

Case	1	
Description	accounting_batch	
Chain/recovery	Chain	
Parent	NA	
PARAM_V1	NA	
PARAM_V2	'000001'	

3.13.6 Input data

Type	Name	Description
Table	ACCOUNTING_SUMMARY_PARAM	
Table	ACCOUNTING_DATA_FILE_SOURCE	

3.13.7 Output data

Type	Name	Description
Table	ACCOUNTING_DATA_FILE_SOURCE	

3.13.8 Warning

The process ends either OK or NOK.

Return code listing: 4527, 4501, 4529, 4524, 4519, 4521, 4999

Batch status query:

```
"SELECT status, ending_status, user_message from pcard_tasks where task_name = 'accounting_batch' and bank_code = $Bank_code; "
```

The trace file query is as follows:

```
"SELECT trace_file_name from pcard_tasks where task_name = 'accounting_batch' and bank_code = $Bank_code; "
```

3.13.9 Recovery procedure

Need to re-run the batch. Transactions that were not processed in the previous batch are included in the current batch.

Blocking	Yes
Clean	No
Mode	Crash
Batch	The same
Parameters	Case 01

3.14 (gen_accounting_file) Batch: Generation accounting file

3.14.1 Batch flow

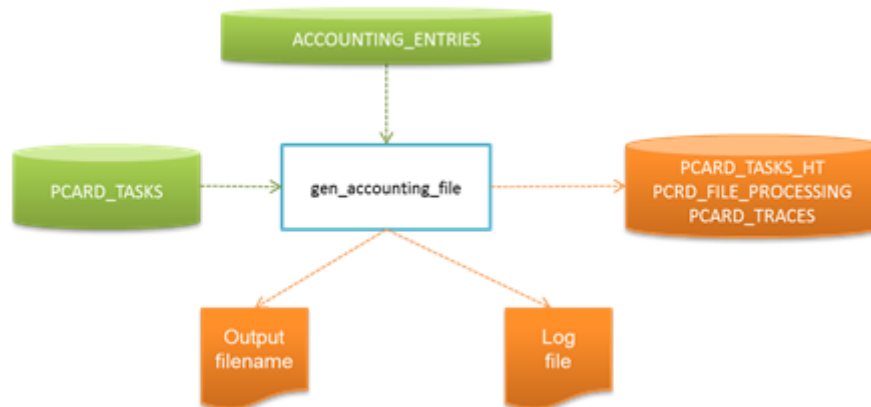


Figure 21: Gen_accounting_file batch

3.14.2 Batch description

This batch generates the file containing the accounting entries generated by **PowerCARD**. It is used to feed external accounting system.

3.14.3 Batch components

Shell	\$SHL/ gen_accounting_file.sh
Binary	\$BIN/gen_accounting_file

Control file	NA
Task	gen_accounting_file
Trace file	\$TRACE/ gen_accounting_file.trc000
Pre Command	Update pcard_tasks Set BUSINESS_DATE = (select LAST_CUTOFF_DATE_BO from switch_cut_off), START_DATE = sysdate , status = 'P' , where task_name = 'gen_accounting_file' and bank_code =\$Bank_code; Commit; Note this script can be replaced by SHL
Supports multi-cutoff	Yes, starting from PowerCARD V3.5
Command	gen_accounting_file.sh <Trace file name><GROUP_CODE>

Default value when the variable <GROUP_CODE> is left blank will be "DFLGRP".

3.14.4 Batch parameters

Below are the fields parametrized in PCARD_TASKS table of this task:

Column name	Parameter	Description	Mandat.
PARAM_V1	Filename	Output filename	Y

3.14.5 Parameter examples

Below are the fields parametrized in PCARD_TASKS table of this task:

Case	1	
Description	Generate the accounting file for external system	
Chain/recovery	Chain	
Parent	accounting_batch	
PARAM_V1	PWCaccounting	

3.14.6 Input data

Type	Name	Description
Table	ACCOUNTING_ENTRIES	

3.14.7 Output data

Type	Name	Description
File	Output file	

3.14.8 Counters

```
----- [ Processed :%d ]
```

The numbers of lines processed are displayed. Header and trailer excluded.

3.14.9 Warning

The process ends either OK or NOK.

Return code listing: 4501, 4519, 4521, 4524, 4526, 4527, 4529, 4590, 4999, 5608

Batch status query:

"SELECT status, ending_status, user_message from pcard_tasks where task_name = 'gen_accounting_file' and bank_code = \$Bank_code; "

The trace file query is as follows:

"SELECT trace_file_name from pcard_tasks where task_name = 'gen_accounting_file' "

3.14.10 Recovery procedure

Need to re-run the batch. Transactions that were not processed in the previous batch are included in the current batch.

Blocking	Yes
Clean	No
Mode	Crash
Batch	The same
Parameters	Case 01

3.15 (processing_ipm_mpe_file) Batch: Load MPE file

3.15.1 Batch flow

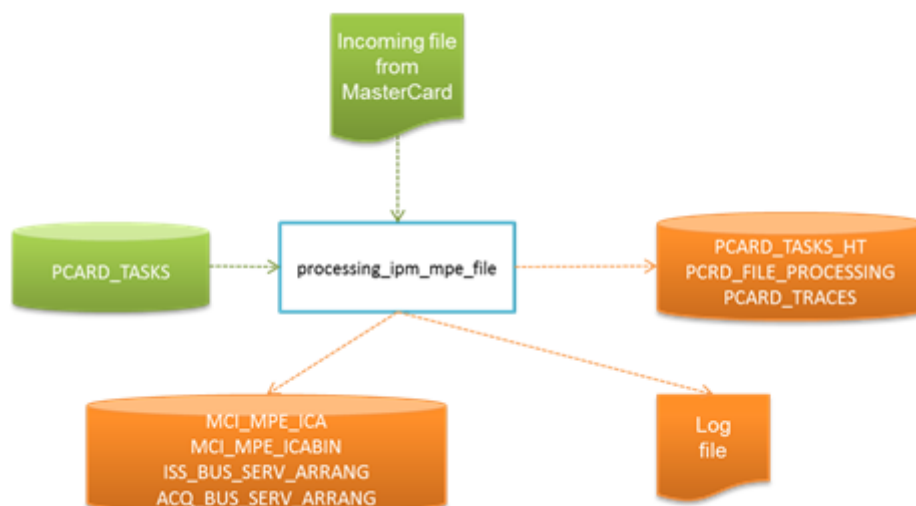


Figure 22: Load MPE file batch

3.15.2 Batch description

The batch is used to load the Integrated Product Messages (IPM) Member Parameter Extract (MPE) files from MasterCard.

3.15.3 Batch components

Shell	\$SHL/processing_ipm_mpe_filesh
Binary	\$BIN/processing_ipm_mpe_file
Control file	N/A
Task	processing_ipm_mpe_file
Trace file	\$TRACE/processing_ipm_mpe_file.TRC000
Command	processing_ipm_mpe_file.sh <Trace file name> <input filename>

3.15.4 Batch parameters

Below are the fields parametrized in PCARD_TASKS table of this task:

Column name	Parameters	Description	Mandat.
BUSINESS_DATE	business_date	Force the processing date to this date [BUSINESS_DATE]	N
PARAM_V1	Input Filename	IPM data in	Y
PARAM_V2	Format	File format. Possible values: EBCDIC1014 or ASCII_1014	N

3.15.5 Parameters example

Below are the fields parametrized in PCARD_TASKS table of this task:

Case	1	
Description	Generate daily and periodic merchant fees	
Chain/recovery	Chain	
Parent	Incoming run from MasterCard	
Child		
PARAM_V1	IPM_MPE	
PARAM_V2	EBCDIC	

3.15.6 Input data

Type	Name	Description
File	Input filename	Input file from MasterCard

3.15.7 Output data

Type	Name	Description
Table	MCI_MPE_ICA	
Table	MCI_MPE_ICABIN	
Table	ISS_BUS_SERV_ARRANG	

Type	Name	Description
Table	ACQ_BUS_SERV_ARRANG	

3.15.8 Counters

```

----- [ Processed :%d ]
----- [ Success :%d ]
----- [ Rejected :%d ]

```

3.15.9 Warning

Batch ends either OK or NOK.

Batch status query:

"SELECT status, ending_status, user_message from pcard_tasks where task_name = 'processing_ipm_mpe_file' and bank_code =\$Bank_code; "

Trace file query:

"SELECT trace_file_name from pcard_tasks where task_name = 'processing_ipm_mpe_file' and bank_code =\$Bank_code; "

Return code listing: 00004, 00016, 00018, 00021, 00021, 99999.

3.15.10 Recovery procedure

Need to rerun the batch. All transactions that were not processed in the previous batch will be included in the current batch.

Blocking	Yes
Clean	No
Mode	Crash
Batch	The same
Parameters	Case 01

3.16 (visa_load_ardef) Batch: VISA ARDEF

3.16.1 Batch flow

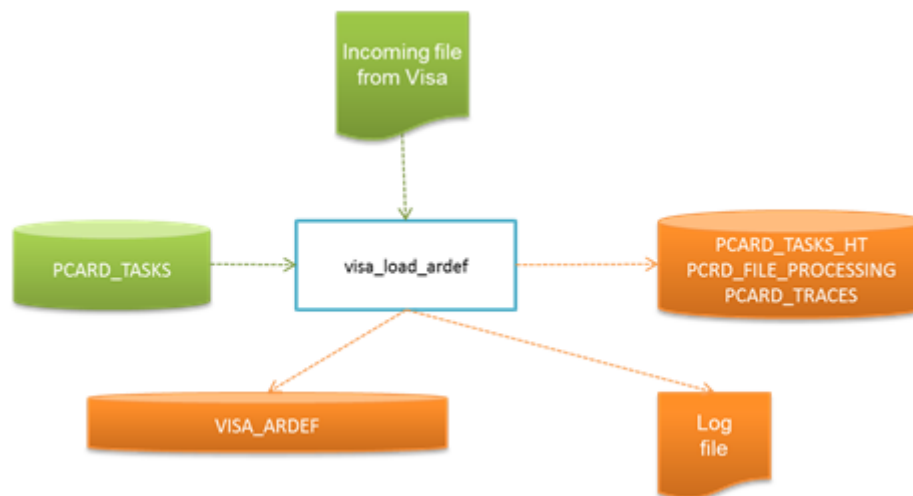


Figure 23: VISA ARDEF batch

3.16.2 Batch description

The batch loads the Visa ARDEF files into PowerCARD.

3.16.3 Batch components

Shell	\$SHL/visa_load_ardef.sh
Binary	\$BIN/visa_load_ardef
Control file	\$CTL/visa_load_ardef.ctl
Task	visa_load_ardef
Trace file	\$TRACE/visa_load_ardef.log
Command	visa_load_ardef.sh <Trace file name> < Input file name>

3.16.4 Batch parameters

Below are the fields parametrized in PCARD_TASKS table of this task:

Column name	Parameter	Description	Mandat.
BUSINESS_DATE	business_date	Force the processing date to this date [BUSINESS_DATE]	N
PARAM_V1	Input Filename	Input file name	O

3.16.5 Parameters example

Below are the fields parametrized in PCARD_TASKS table of this task:

Case	1	
Description	Generate daily and periodic merchant fees	

Chain/recovery	Chain	
Parent	Incoming tun from Visa	
Child		
PARAM_V1	VISA_ARDEF	

3.16.6 Input data

Type	Name	Description
File	Input file from Edit package	ARDEF table

3.16.7 Output data

Type	Name	Description
Table	VISA_ARDEF	

3.16.8 Counters

```

----- [ Processed :%d ]
----- [ Success :%d ]
----- [ Rejected :%d ]

```

3.16.9 Warning

Batch ends either OK or NOK.

Batch status query:

"SELECT status, ending_status, user_message from pcard_tasks where task_name = 'visa_load_ardef' and bank_code =\$Bank_code; "

Trace file query:

"SELECT trace_file_name from pcard_tasks where task_name = 'visa_load_ardef' and bank_code =\$Bank_code; "

Return code listing: 00016, 00018, 00017,00021, 99999.

3.16.10 Recovery procedure

Need to rerun the batch. All transactions that were not processed in the previous batch will be included in the current batch.

Blocking	Yes
Clean	No
Mode	Crash
Batch	The same
Parameters	Case 01

3.17 (visa_load_bin) Batch: VISA BIN

3.17.1 Batch flow

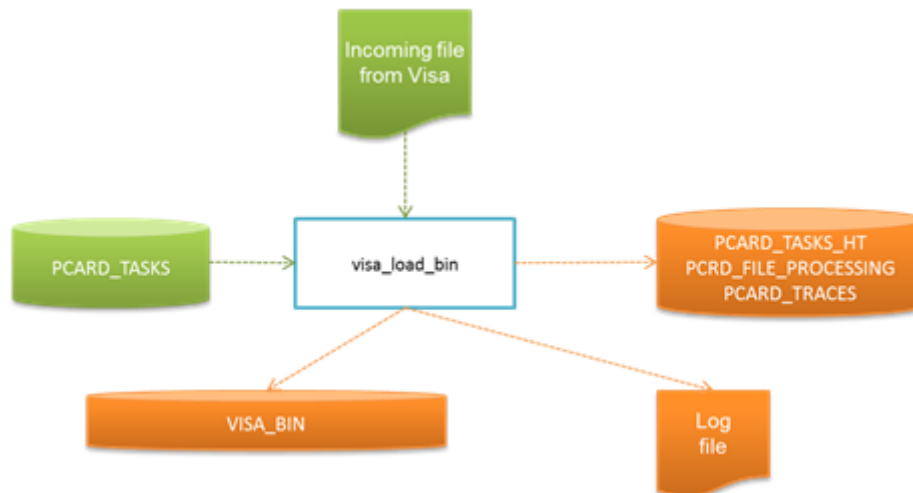


Figure 24: VISA BIN batch

3.17.2 Batch description

The batch loads the Visa BIN tables into PowerCARD.

3.17.3 Batch components

Shell	\$SHL/visa_load_bin.sh
Binary	\$BIN/visa_load_bin
Control file	\$CTL/visa_load_bin.ctf
Task	visa_load_bin
Trace file	\$TRACE/visa_load_bin.log
Command	visa_load_bin.sh <Trace file name> < Input file name>

3.17.4 Batch parameters

Below are the fields parametrized in PCARD_TASKS table of this task:

Column name	Parameter	Description	Mandat.
BUSINESS_DATE	business_date	Force the processing date to this date [BUSINESS_DATE]	N
PARAM_V1	Input Filename	Input file name	O
PARAM_V2	Record length	Length of records in file (81)	O

3.17.5 Parameters example

Below are the fields parametrized in PCARD_TASKS table of this task:

Case	1	
------	---	--

Description	Generate daily and periodic merchant fees	
Chain/recovery	Chain	
Parent	Incoming run from Visa	
Child		
PARAM_V1	VISA_BIN	
PARAM_V2	81	

3.17.6 Input data

Type	Name	Description
File	VISA BIN file	File to be extracted from Edit package

3.17.7 Output data

Type	Name	Description
Table	VISA_BIN	

3.17.8 Counters

```

----- [ Processed :%d ]
----- [ Success :%d ]
----- [ Rejected :%d ]

```

3.17.9 Warning

Batch ends either OK or NOK.

Batch status query:

"SELECT status, ending_status, user_message from pcard_tasks where task_name = 'visa_load_bin' and bank_code = \$Bank_code; "

Trace file query:

"SELECT trace_file_name from pcard_tasks where task_name = 'visa_load_bin' and bank_code = \$Bank_code; "

Return code listing: 00016, 00018, 00017, 99999

3.17.10 Recovery procedure

Need to rerun the batch. All transactions that were not processed in the previous batch will be included in the current batch.

Blocking	Yes
Clean	No
Mode	Crash
Batch	The same
Parameters	Case 01

3.18 (visa_load_plus) Batch: VISA PLUS

3.18.1 Batch flow

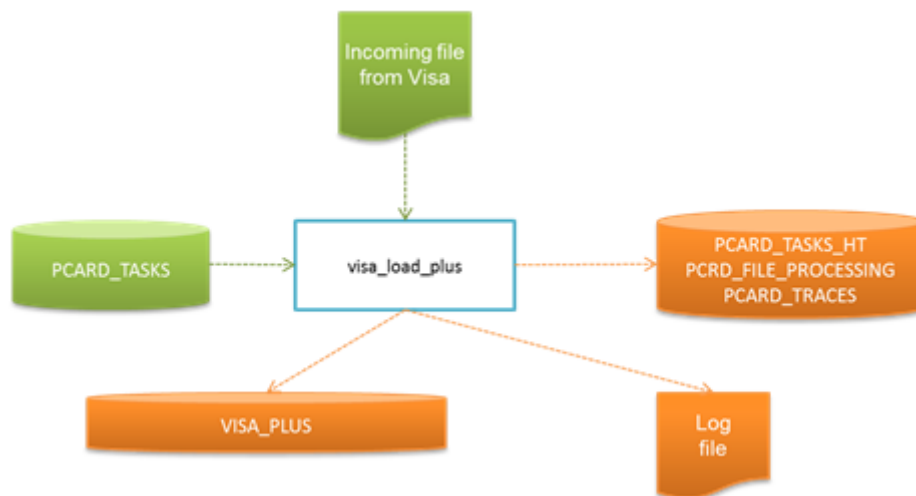


Figure 25: VISA PLUS batch

3.18.2 Batch description

The batch loads the Visa PLUS files into PowerCARD. This batch applies only for the Caribbean release.

3.18.3 Batch components

Shell	\$SHL/visa_load_plus.sh
Binary	\$BIN/visa_load_plus
Control file	N/A
Task	visa_load_plus
Trace file	\$TRACE/visa_load_plus.log
Command	visa_load_plus.sh <Trace file name> < Input file name>

3.18.4 Batch parameters

Below are the fields parametrized in PCARD_TASKS table of this task:

Column name	Parameter	Description	Mandat.
BUSINESS_DATE	business_date	Force the processing date to this date [BUSINESS_DATE]	N
PARAM_V1	Input Filename	Input file name	O

3.18.5 Parameters example

Below are the fields parametrized in PCARD_TASKS table of this task:

Case	1	
------	---	--

Description	Generate daily and periodic merchant fees	
Chain/recovery	Chain	
Parent	Incoming run from Visa	
Child		
PARAM_V1	VISA_PLUS	

3.18.6 Input data

Type	Name	Description
File	Bin PLUS file	File to be extracted from the Edit Package

3.18.7 Output data

Type	Name	Description
Table	VISA_PLUS	

3.18.8 Counters

```

----- [ Processed :%d ]
----- [ Success :%d ]
----- [ Rejected :%d ]

```

3.18.9 Warning

Batch ends either OK or NOK.

Batch status query:

"SELECT status, ending_status, user_message from pcard_tasks where task_name = 'visa_load_plus' and bank_code =\$Bank_code; "

Trace file query:

"SELECT trace_file_name from pcard_tasks where task_name = 'visa_load_plus' and bank_code =\$Bank_code; "

Return code listing: 00004, 00016, 00017, 00018, 00019, 00021, 00024, 00025, 00026, 00027, 00028, 00043, 00054, 00055, 99999

3.18.10 Recovery procedure

Need to rerun the batch. All transactions that were not processed in the previous batch will be included in the current batch.

Blocking	Yes
Clean	No
Mode	Crash
Batch	The same
Parameters	Case 01

4 Annexe A Tables Structures

4.1 PCARD_TASKS

COL ID	DATA ELEMENT	DESCRIPTION /COMMENTS	FORMAT	LEN
001	BANK_CODE	NOT NULL	CHAR	6
002	TASK_NAME	NOT NULL	VARCHAR2	60
003	TASK_WORDING	task_wording IS NOT NULL	VARCHAR2	80
004	TASK_PERIOD	NVL(task_period,'?') IN ('DA', 'MO', 'QA', 'HY', 'YE', 'WK', 'UR', 'FN')	CHAR	2
005	PRIORITY_LIST	priority_list IS NOT NULL	NUMBER	22
006	TASK_GROUP_CODE		VARCHAR2	6
007	TRACE_FILE_NAME	trace_file_name IS NOT NULL	VARCHAR2	256
008	PREVIOUS_TASK_NAME		VARCHAR2	60
009	CHECK_PREVIOUS_TASK_STATUS	check_previous_task_status in ('Y','N')	CHAR	1
010	CHECK_PREVIOUS_TASK_BDATE	check_previous_task_bdate in ('Y','N')	CHAR	1
011	NEXT_TASK_NAME		VARCHAR2	60
012	START_AUTO_NEXT_TASK	start_auto_next_task in ('C','A','Q','N')	CHAR	1
013	EXCLUSIVE_TASK		VARCHAR2	60
014	TASK_CHAIN_CODE		VARCHAR2	6
015	ORDER_SEQUENCE_IN_CHAIN		NUMBER	22
016	TASK_CHAIN_SEQ		NUMBER	22
017	STATUS	status in ('R','P','S','U','X')	CHAR	1
018	START_DATE		DATE	7
019	END_DATE		DATE	7
020	ENDING_STATUS		NUMBER	22
021	USER_NAME		VARCHAR2	15
022	USER_MESSAGE		VARCHAR2	512
023	ORA_ERR		NUMBER	22
024	BUSINESS_DATE		DATE	7
025	PARAMETER_V1_OPTION	parameter_v1_option in ('N','O','M')	CHAR	1
026	PARAMETER_V1_TYPE	parameter_v1_type in ('D','V','N')	CHAR	1
027	PARAMETER_V1_LMIN		NUMBER	3
028	PARAMETER_V1_LMAX		NUMBER	3
029	PARAMETER_V1_LABEL		VARCHAR2	40
030	PARAMETER_V1_DATA		VARCHAR2	256
031	PARAMETER_V2_OPTION	parameter_v2_option in ('N','O','M')	CHAR	1
032	PARAMETER_V2_TYPE	parameter_v2_type in ('D','V','N')	CHAR	1
033	PARAMETER_V2_LMIN		NUMBER	3

COL ID	DATA ELEMENT	DESCRIPTION /COMMENTS	FORMAT	LEN
034	PARAMETER_V2_LMAX		NUMBER	3
035	PARAMETER_V2_LABEL		VARCHAR2	40
036	PARAMETER_V2_DATA		VARCHAR2	256
037	PARAMETER_V3_OPTION	parameter_v3_option in ('N','O','M')	CHAR	1
038	PARAMETER_V3_TYPE	parameter_v3_type in ('D','V','N')	CHAR	1
039	PARAMETER_V3_LMIN		NUMBER	3
040	PARAMETER_V3_LMAX		NUMBER	3
041	PARAMETER_V3_LABEL		VARCHAR2	40
042	PARAMETER_V3_DATA		VARCHAR2	256
043	PARAMETER_V4_OPTION	parameter_v4_option in ('N','O','M')	CHAR	1
044	PARAMETER_V4_TYPE	parameter_v4_type in ('D','V','N')	CHAR	1
045	PARAMETER_V4_LMIN		NUMBER	3
046	PARAMETER_V4_LMAX		NUMBER	3
047	PARAMETER_V4_LABEL		VARCHAR2	40
048	PARAMETER_V4_DATA		VARCHAR2	256
049	PARAMETER_V5_DATA		VARCHAR2	256
050	PARAMETER_V5_OPTION		CHAR	1
051	PARAMETER_V5_TYPE		CHAR	1
052	PARAMETER_V5_LMIN		NUMBER	3
053	PARAMETER_V5_LMAX		NUMBER	3
054	PARAMETER_V5_LABEL		VARCHAR2	40
055	PARAMETER_V6_DATA		VARCHAR2	256
056	PARAMETER_V6_OPTION		CHAR	1
057	PARAMETER_V6_TYPE		CHAR	1
058	PARAMETER_V6_LMIN		NUMBER	3
059	PARAMETER_V6_LMAX		NUMBER	3
060	PARAMETER_V6_LABEL		VARCHAR2	40
061	PARAMETER_V7_DATA		VARCHAR2	256
062	PARAMETER_V7_OPTION		CHAR	1
063	PARAMETER_V7_TYPE		CHAR	1
064	PARAMETER_V7_LMIN		NUMBER	3
065	PARAMETER_V7_LMAX		NUMBER	3
066	PARAMETER_V7_LABEL		VARCHAR2	40
067	PARAMETER_V8_DATA		VARCHAR2	256
068	PARAMETER_V8_OPTION		CHAR	1
069	PARAMETER_V8_TYPE		CHAR	1
070	PARAMETER_V8_LMIN		NUMBER	3

COL ID	DATA ELEMENT	DESCRIPTION /COMMENTS	FORMAT	LEN
071	PARAMETER_V8_LMAX		NUMBER	3
072	PARAMETER_V8_LABEL		VARCHAR2	40
073	PARAMETER_V9_DATA		VARCHAR2	256
074	PARAMETER_V9_OPTION		CHAR	1
075	PARAMETER_V9_TYPE		CHAR	1
076	PARAMETER_V9_LMIN		NUMBER	3
077	PARAMETER_V9_LMAX		NUMBER	3
078	PARAMETER_V9_LABEL		VARCHAR2	40
079	PARAMETER_V10_DATA		VARCHAR2	256
080	PARAMETER_V10_OPTION		CHAR	1
081	PARAMETER_V10_TYPE		CHAR	1
082	PARAMETER_V10_LMIN		NUMBER	22
083	PARAMETER_V10_LMAX		NUMBER	22
084	PARAMETER_V10_LABEL		VARCHAR2	40
085	COUNTER_1		NUMBER	22
086	COUNTER_2		NUMBER	22
087	COUNTER_3		NUMBER	22
088	COUNTER_4		NUMBER	22
089	COUNTER_5		NUMBER	22
090	AMOUNT_1		NUMBER	22
091	AMOUNT_2		NUMBER	22
092	AMOUNT_3		NUMBER	22
093	AMOUNT_4		NUMBER	22
094	AMOUNT_5		NUMBER	22
095	SENSITIVE_OPERATION_RECORD		CHAR	1
096	USER_CREATE		VARCHAR2	15
097	DATE_CREATE		DATE	7
098	USER_MODIF		VARCHAR2	15
099	DATE_MODIF		DATE	7

4.2 PCARD_TRACES

COL ID	DATA ELEMENT	DESCRIPTION / COMMENTS	FORMAT	LEN
001	USER_TRACE		VARCHAR2	15
002	DATE_TRACE		DATE	7
003	TYPE_OF_TRACE		CHAR	1
004	SEVERITY_LEVEL		CHAR	1
005	MODULE_CODE		VARCHAR2	3

COL ID	DATA ELEMENT	DESCRIPTION / COMMENTS	FORMAT	LEN
006	MESSAGE_NUMBER		VARCHAR2	4
007	PACKAGE_NAME		VARCHAR2	30
008	FUNCTION_NAME		VARCHAR2	30
009	SPY_COUNTER		NUMBER	22
010	USER_MESSAGE		VARCHAR2	4000
011	ORA_ERR		NUMBER	22
012	BUSINESS_DATE		DATE	7
013	ERROR_CODE		VARCHAR2	9
014	ORA_ERROR_CODE		VARCHAR2	9
015	BANK_CODE		VARCHAR2	6
016	TASK_NAME		VARCHAR2	50
017	TASK_ID		VARCHAR2	10
018	JOB_NUMBER		NUMBER	22
019	USER_CREATE		VARCHAR2	15

4.3 PCARD_ERRORS

COL ID	DATA ELEMENT	DESCRIPTION / COMMENTS	FORMAT	LEN
001	ERROR_SEQUENCE	NOT NULL LA SÉQUENCE DE L'ERREUR	VARCHAR2	20
002	ERROR_CODE	CODE PowerCARD DE L'ERREUR	VARCHAR2	10
003	ORA_ERROR_CODE	CODE ORACLE DE L'ERREUR	VARCHAR2	10
004	BANK_CODE	BANQUE	VARCHAR2	6
005	TASK_NAME	NOM DU BATCH	VARCHAR2	50
006	TASK_ID	IDENTIFIANT DU BATCH	VARCHAR2	10
007	JOB_NUMBER	NUMÉRO DU JOB QUI A CAUSE L'ERREUR	NUMBER	22
008	MESSAGE	MESSAGE DE L'ERREUR	VARCHAR2	4000
009	PACKAGE_NAME	PACKAGE OÙ IL Y'A L'ERREUR	VARCHAR2	100
010	FUNCTION_NAME	FONCTION OÙ IL	VARCHAR2	100

COL ID	DATA ELEMENT	DESCRIPTION / COMMENTS	FORMAT	LEN
		Y'A L'ERREUR		
011	MODULE_CODE	MODULE DE L'ERREUR	VARCHAR2	3
012	STEP_TASK_FUNCTION		VARCHAR2	100
013	BUSINESS_DATE	LE BUSINESS DATE	DATE	7
014	USER_CREATE	UTILISATEUR RESPONSABLE DE LA TRACE	VARCHAR2	15
015	DATE_CREATE	DATE TRACE DE L'ERREUR	DATE	7

4.4 PCARD_MESSAGE_APP

COL ID	DATA ELEMENT	DESCRIPTION / COMMENTS	FORMAT	LEN
001	APP_CODE	le préfixe du code de l'erreur 'PWC'	CHAR	3
002	ERROR_CODE	Le code de l'erreur	VARCHAR2	10
003	ERROR_NAME	Le nom de l'erreur	VARCHAR2	20
004	LANGUAGE_CODE	La langue du message	CHAR	3
005	MESSAGE	Le message d'erreur	VARCHAR2	200
006	ACTION	L'action de correction de l'erreur	VARCHAR2	200

4.5 RESOURCES

COL ID	DATA ELEMENT	DESCRIPTION / COMMENTS	FORMAT	LEN
001	RESOURCE_ID	NOT NULL	CHAR	6
002	RESOURCE_CODE		CHAR	2
003	ABRV_WORDING	abrv_wording IS NOT NULL	VARCHAR2	16
004	WORDING	wording IS NOT NULL	VARCHAR2	32
005	COMPONENT_TYPE	NVL(component_type, '?') IN ('PRIS', 'IPC', 'UFS', 'DBTS')	VARCHAR2	4
006	RESOURCE_TYPE	NVL(resource_type, '?') IN ('G', 'N', 'S', 'P', 'A', 'B', 'H', 'V', 'I')	CHAR	1
007	STATUS	NVL (status, '?') IN ('N', 'U')	CHAR	1

COL ID	DATA ELEMENT	DESCRIPTION /COMMENTS	FORMAT	LEN
008	SUBJECT_OF_SUPERVISE	NVL (subject_of_supervise, '?') IN ('N','Y')	CHAR	1
009	OBJECT_NAME		VARCHAR2	60
010	PRIS_OBJECT_PATH		VARCHAR2	60
011	IPC_CRITICAL_LEVEL_NB_MSG		NUMBER	22
012	IPC_CRITICAL_PERCENT_FREESPACE		NUMBER	22
013	UFS_CRITICAL_PERCENT_FREESPACE		NUMBER	22
014	UFS_CRITICAL_SIZE_FREESPACE		NUMBER	22
015	DBTS_CRITICAL_PERC_FREESPACE		NUMBER	22
016	PRIS_START_SCRIPT_NAME		VARCHAR2	40
017	PRIS_START_SCRIPT_PATH		VARCHAR2	60
018	PRIS_STOP_SCRIPT_NAME		VARCHAR2	40
019	PRIS_STOP_SCRIPT_PATH		VARCHAR2	60
020	PRIS_SUBJECT_OF_AUTO_START	NVL (pris_subject_of_auto_start, '?') IN ('N', 'Y')	CHAR	1
021	PRIS_SUPPORT_ADM_MESSAGE	NVL (pris_support_adm_message, '?') IN('N', 'Y')	CHAR	1
022	PRIS_COMMS_ID		CHAR	2
023	PRIS_PROTOCOL_ID		CHAR	2
024	PRIS_APPLICATION_RELEASE		CHAR	3
025	PRIS_BANK_CODE		CHAR	6
026	PRIS_ACQ_TRANSPORT_KEY_NUMBER		CHAR	3
027	PRIS_ISS_TRANSPORT_KEY_NUMBER		CHAR	3
028	PRIS_MAC_KEY_NUMBER		CHAR	3
029	PRIS_MASTER_KEY_NUMBER		CHAR	3
030	PRIS_RESOURCE_AUTHO_FLAG	NVL (pris_resource_autho_flag, '?') IN ('Y', 'N')	CHAR	1
031	PRIS_CONNECT_MODE	NVL (pris_connect_mode, '?') IN ('S', 'M')	CHAR	1
032	PRIS_DEVICE_NAME		VARCHAR2	64
033	PRIS_REMOTE_APPLICATION_NAME		VARCHAR2	64
034	PRIS_REMOTE_APPLICATION_ID		VARCHAR2	64
035	PRIS_LOCAL_APPLICATION_NAME		VARCHAR2	64
036	PRIS_LOCAL_APPLICATION_ID		VARCHAR2	64
037	PRIS_LOG_FILE_NAME		VARCHAR2	20
038	PRIS_LOG_MAX_SIZE		NUMBER	22
039	PRIS_START_CYCLE		NUMBER	22
040	PRIS_END_CYCLE		NUMBER	22
041	PRIS_TRACE_LEVEL	NVL (pris_trace_level, '?') IN ('0', '1', '2', '3', '4', '5', '6')	CHAR	1

COL ID	DATA ELEMENT	DESCRIPTION /COMMENTS	FORMAT	LEN
042	PRIS_NB_STATIC_THREADS		NUMBER	22
043	PRIS_NB_USER_THREADS_VS_KERNEL		NUMBER	22
044	PRIS_TIMER_ACQUIRER		NUMBER	22
045	PRIS_DELTA_TIMER_ISS_ACQ		NUMBER	22
046	PRIS_ECHO_TEST_TIMER		NUMBER	22
047	PRIS_TRANSACTION_SIM_MAX		NUMBER	22
048	PRIS_GEN_REVERSAL_TIMEOUT_LATE	NVL (pris_gen_reversal_timeout_late, '?') IN ('T', 'L')	CHAR	1
049	PRIS_MIN_TROUBLES_IND		NUMBER	22
050	PRIS_MAX_TROUBLES_IND		NUMBER	22
051	PRIS_CURRENT_CYCLE		NUMBER	22
052	RESOURCE_LIVE	NVL (resource_live, '?') IN ('Y', 'C', 'N')	CHAR	1
053	PRIS_PROCESSING_STEP	NVL (pris_processing_step, '?') IN ('OFF', 'OPN', 'CNT', 'SAF', 'OFL', 'ONL')	CHAR	3
054	PRIS_INTERNAL_ID		VARCHAR2	32
055	PRIS_PID		NUMBER	22
056	PRIS_PPID		NUMBER	22
057	PRIS_TROUBLES_IND	NVL (pris_troubles_ind, '?') IN ('Y', 'N')	CHAR	1
058	PRIS_TROUBLES_IND_UPD_RESID		CHAR	2
059	PRIS_TROUBLES_IND_UPDATE_DATE		DATE	7
060	SENSITIVE_OPERATION_RECORD	NVL (sensitive_operation_record, '?') IN ('I', 'M', 'D', 'X')	CHAR	1
061	USER_CREATE		VARCHAR2	15
062	DATE_CREATE		DATE	7
063	USER_MODIF		VARCHAR2	15
064	DATE_MODIF		DATE	7

4.6 PCRD_FILE_PROCESSING

COL ID	DATA ELEMENT	DESCRIPTION / COMMENTS	FORMAT	LEN
1	BANK_CODE	NOT NULL	CHAR	6
2	PHYSICAL_FILE_NAME	NOT NULL	VARCHAR2	80
3	SEQUENCE_NUMBER	NOT NULL	NUMBER	12
4	CURRENCY_CODE		CHAR	3
5	SENDER_NAME		VARCHAR2	30
6	INTERNAL_FILE_NAME		VARCHAR2	40
7	INTERNAL_SEQUENCE_FILE		VARCHAR2	20

COL ID	DATA ELEMENT	DESCRIPTION / COMMENTS	FORMAT	LEN
8	FIRST_RECORD_FILE		VARCHAR2	2048
9	INTERNAL_CREATION_DATE		DATE	7
10	PROCESSING_START_DATE		DATE	7
11	PROCESSING_END_DATE		DATE	7
12	PROCESSING_STATUS		CHAR	2
13	FILE_NUMBER_RECORDS		NUMBER	12
14	PROCESSING_NUMBER_RECORDS		NUMBER	12
15	NUMBER_VALID_RECORDS		NUMBER	12
16	NUMBER_REJECT_RECORDS		NUMBER	12
17	NUMBER_PROCESSED_CANCELLED		NUMBER	12
18	GLOBAL_REJECT_FILE		CHAR	1
19	GLOBAL_REJECT_REASON		VARCHAR2	100
20	BANK_AMOUNT		NUMBER	18
21	MERCHANT_AMOUNT		NUMBER	18
22	COMMISSION_AMOUNT		NUMBER	18
23	SUM_COMMISSION_NOT_REVERSED		NUMBER	18
24	FILE_COMMISSION_REVERSAL_TRX		NUMBER	18
25	FILE_COMMISSION_SUSPEND_TRX		NUMBER	18
26	FILE_COMMISSION_NORMAL_TRX		NUMBER	18
27	SUM_TRX_CD_VOUCHER_VALID		NUMBER	18
28	SUM_TRX_PURCHASE_VALID		NUMBER	18
29	SUM_TRX_REVERSAL_VALID		NUMBER	18
30	SUM_TRX_CANCELLED_VALID		NUMBER	18
31	SUM_TRX_PURCHASE_AMOUNT		NUMBER	18
32	SUM_TRX_REVERSAL_AMOUNT		NUMBER	18
33	SUM_TRX_CANCEL_AMOUNT		NUMBER	18
34	DIFF_PURCHASE_PROCESS_FILE		NUMBER	18
35	DIFF_REVERSAL_PROCESS_FILE		NUMBER	18
36	DIFF_CANCELLED_PROCESS_FILE		NUMBER	18
37	SIGN_PURCHASE_PROCESS_FILE		CHAR	1
38	SIGN_REVERSAL_PROCESS_FILE		CHAR	1
39	SIGN_CANCELLED_PROCESS_FILE		CHAR	1
40	NUMBER_PURCHASE_TRX		NUMBER	12
41	NUMBER_REVERSAL_TRX		NUMBER	12
42	NUMBER_CANCEL_TRX		NUMBER	12
43	FILE_NBR_PURCHASE_TRX		NUMBER	12
44	FILE_NBR_REVERSAL_TRX		NUMBER	12

COL ID	DATA ELEMENT	DESCRIPTION / COMMENTS	FORMAT	LEN
45	FILE_NBR_CANCEL_TRX		NUMBER	12
46	DIFF_BANK_MERCHANT_AMOUNT		NUMBER	18
47	SIGN_BANK_MERCHANT_AMOUNT		CHAR	1
48	DIFF_MERCHANT_TRANS_AMOUNT		NUMBER	18
49	SIGN_MERCHANT_TRANS_AMOUNT		CHAR	1
50	PRINT_FLAG_LETTER		CHAR	1
51	PRINT_FLAG_REPORT		CHAR	1
52	MEMO_1		VARCHAR2	512
53	MEMO_2		VARCHAR2	512
54	MEMO_3		VARCHAR2	512
55	FILE_TYPE		CHAR	1
56	RECORD_FLAG		CHAR	1
57	LETTER_FLAG		CHAR	1
58	SERVICE_CODE_SENDING_FILE		VARCHAR2	15
59	CLEARING_BANK_CODE		CHAR	6
60	PROCESSING_DATE		DATE	7
61	USER_CREATE		VARCHAR2	15
62	DATE_CREATE		DATE	7
63	USER_MODIF		VARCHAR2	15
64	DATE_MODIF		DATE	7

5 Annexe B Batch return code

Code	Message
00000	BATCH %1 SUCCESSFULLY ENDED
00001	YOU TRIED TO PERFORM AN INSERT STATEMENT THAT CREATED A DOUBLE VALUE IN TABLE %1 FOR PARAMETERS %2
00002	YOU WERE WAITING FOR A RESOURCE AND YOU TIMED OUT.
00003	YOU TRIED TO USE A CURSOR THAT DOES NOT EXIST. THIS CAN OCCUR BECAUSE YOU HAVE EXECUTED A CURSOR FETCH OR CLOSE BEFORE OPENING THE CURSOR
00004	A SELECT STATEMENT HAS BEEN EXECUTED FROM A TABLE (%1) AND NO RECORDS HAVE BEEN RETURNED (%2)
00005	YOU TRIED TO EXECUTE A SELECT STATEMENT INTO TABLE %1 USING PARAMETERS %2 AND MORE THAN ONE RECORD HAS BEEN RETURNED.
00006	YOU TRIED TO DIVIDE A NUMBER BY ZERO.
00007	YOU TRIED TO EXECUTE AN SQL STATEMENT THAT TRIED TO CONVERT A STRING TO NUMBER BUT IT FAILED.
00008	NOT ENOUGH MEMORY OR CORRUPTED.
00009	YOU TRIED TO MAKE AN OPERATION AND THERE WAS AN ERROR IN CONVERTING THE TRUNCATION, THE NUMERIC DATA OR CHARACTERS ARE INVALID.
00010	YOU TRIED TO OPEN A CURSOR THAT IS ALREADY OPEN.
00011	YOUR PROGRAM TRIES TO ASSIGN VALUES TO NON INITIALIZED ATTRIBUTES.
00012	CURSOR VARIABLE AND PL /SQL CURSOR VARIABLE INVOLVED HAVE INCOMPATIBLE RETURN TYPES
00013	CONVERSION OF A STRING IN A ROWID FAILS BECAUSE THE STRING DOES NOT REPRESENT A VALID ROWID.
00014	NONE OF CHOICE IN THE CLAUSE "WHEN" OF A "CASE" INSTRUCTION IS SELECTED, AND THERE IS NO "ELSE" CLAUSE.
00015	THE JOINT BETWEEN THE TABLES 1% RETURNS NO RECORDS
00016	AN EXCEPTION OCCURS DURING INSERT INTO THE TABLE %1 FOR THE PARAMETERS %2
00017	AN EXCEPTION OCCURS WHEN UPDATE TABLE %1 FOR THE PARAMETERS %2
00018	AN EXCEPTION OCCURS WHEN DELETING FROM TABLE 1% FOR THE PARAMETERS %2
00019	YOU TRIED TO EXECUTE AN UPDATE STATEMENT THAT RETURNED MORE THAN ONE RECORDS IN THE TABLE 1% FOR THE PARAMETERS %2

Code	Message
00020	YOU TRIED TO EXECUTE A DELETE STATEMENT THAT RETURNED MORE THAN ONE RECORDS IN THE TABLE 1% FOR THE PARAMETERS %2
00021	AN EXCEPTION OCCURS IN A SELECT STATEMENT FROM TABLE 1% FOR THE PARAMETERS 2%
00022	THE INFORMATION SELECTED FROM TABLE 1% ARE INCONSISTENT WITH THE HISTORY OF THE TABLE 2%
00023	THERE IS MORE THAN ONE JOB THAT ENDS UNSUCCESSFULLY FOR BATCH 1%
00024	THERE IS MORE THAN ONE JOB THAT ENDS UNSUCCESSFULLY AND THE NUMBER OF PROCESSED RECORDS IS LOWER THAN THE TOTAL NUMBER OF RECORDS FOR 1%
00025	DATA ARE INCONSISTENT FOR THE PARAMETERS ==>%1, % 2
00026	AN ERROR OCCURS IN THE THE JOB: 1%
00027	AN ERROR OCCURS AT THE VALUE:%1
00028	AN ERROR OCCURS IN EXECUTING THE FOLLOWING QUERY : %1
00029	EXTRACTING THE VALUE %1 FROM STRING %2 HAS FAIL
00030	THE UPDATE ON TABLE %1 FOR THE %2 PARAMETERS HAS MIT UPDATE NO LINE.
00031	THE ALLOCATION OF %1 IN THE %2 HAS AN ERROR
00040	YOU TRIED TO EXECUTE AN INSERT STATEMENT FOR UPDATE IN TABLE %1 FOR THE %2 PARAMETERS
00041	ONE (OR SEVERAL) ARGUMENTS REQUIRED (S) IS (ARE) NULL, %1
00042	THE PARAMETER %1 IS UNDEFINED, CHECK THE %2 PARAMETER %3 TABLES
00043	THE VARIABLE GLOBAL %1 IS NULL
00044	DATA FROM THE %1 SETTINGS ARE NULL
00045	UNABLE TO GENERATE THE NEXT VALUE IN THE SEQUENCE %1
00046	AT LEAST ONE OF THE PARAMETERS %1 MUST BE FILLED
00047	YOU RUN A SELECT COUNT(*) STATEMENT IN TABLE %1 WITH %2 PARAMETERS AND NO LINE WAS RETURNED
00048	%1 ENVIRONMENT VARIABLE IS NOT SET
00049	THE %1 PARAMETER ERROR
00050	CONVERSION MODE IS BAD %1
00051	THE RATE CATEGORY IS BAD %1
00052	THE CURRENCY IS NOT ACCEPTED - STATUS = 1

Code	Message
00053	MORE THAN A JOB ARE ENDED WITHOUT SUCCESS FOR THE %1 SETTINGS
00054	TYPE %1 NOT SUPPORTED ROWID
00055	TYPE UNKNOWN %1
00056	DIRECTORY VFILE INVALID NAME
00057	INVALID MODE
00058	OPERATION INVALID
00059	THE MAXLINESIZE IS INVALID
00060	ERROR IN THE STORYTELLER OF THE LOOP > %1
00061	NUMBER OF LOOP WILL EXCEED THE MAX OF TASKS > %1
00062	ATTENTION: RECURSIVE ON THE NEXT LOOP TASK %1
00063	INVALID EXCEPTION PATH. PLEASE CHECK THE VARIABLE 1
00064	ERROR READING FROM THE EXCEPTION
00065	ERROR CONVERTING %1
00066	ERROR WHEN UPDATE VDELETE OF TABLE %1 WITH %2 PARAMETERS
00067	NO RECORD GENERATED FOR FILES OF COLLECTION %1
00072	ERROR RECOVERY IN THE CURRENCYRATE FOR THE PARAMETERS FOLLOWING: %1
00073	MATCHING WITH THE TABLE ERROR: %1
00074	INCOMPATIBILITY BETWEEN %1 AND %2
00075	UNABLE TO SET THE ORIGIN CODE FOR THE %1 SETTINGS
00077	RAISED WHEN NO MORE DATA IS AVAILABLE TO READ FROM THE HSM CONNECTION
00078	NO DATA IS READ FROM THE HSM AND A READ TIMEOUT OCCURRED
00506	ERROR OF AUDITS OF THE DAYS HOLIDAYS
01011	BATCH %1 MUST BE STARTED AFTER THE LAUNCH OF THE %2 BATCH
01111	BATCH %1 COMPLETED PARTIALLY SUCCESSFUL
01168	NUMBER OF TRANSACTIONS NUMBER OF RECYCLED <>TRANSACTIONS NUMBER OF TRANSACTIONS NUMBER OF VALID TRANSACTIONS REJECTED CAPTURED
01169	ERROR READING FROM THE CUTOFF FOR THE %1 BATCH SEQUENCE
01170	ERROR RECOVERY OF THE COST OF MANAGING TRANSACTIONS ISSUERVACQUIRER

Code	Message
01173	PROBLEM INITIALIZING OF THE STORYTELLERS FOR BATCH %1
01174	ERROR GETTING STATUS FOR TASK %1
01175	ERROR IN THE MAIN FUNCTION IN BATCH %1
01176	TRUNCATE THE TABLES OF THE CRE ERROR
01260	TRUNCATE THE TABLES OF REJECTED TRANSACTIONS ERROR
01261	TRUNCATE COMMITTED TRANSACTIONS TABLES ERROR
01263	ERROR INITIALIZING THE AUTHO_ACTIVITY_MTH TABLE
01264	ERROR INITAILISATION OF GLOBAL VARIABLES
01265	ERROR OF UPDATE OF THE STATUS OF THE BATCH %1 IN THE %2 TABLE
01266	THE STORYTELLERS OF THE BATCH CALCULATION ERROR
01267	ERROR OF TRANSFER FROM THE TEMPORARY TABLE TO RELEASE TO THE TABLE PCARD_CRE_REM_ACQUITTEMENT
01268	ERROR OF TRANSFER FROM THE TEMPORARY TABLE TO RELEASE TO THE TABLE CRE_REM_CONTROL
02905	PARAMETERS VERIFICATION ERROR
02915	ERROR SETTINGS NFCCA INITIALISATION
02916	ERROR CONVERTING FILE TO NFCCA
02925	NOT FOUND COMPENSATION PARTNER
02926	ERROR CONVERTING THE SUMMER DELIVERED TO NFCCA
02935	THE HEADER CONTAINS ERRORS
02945	FILE PROCESSING ERROR
02955	FILE NFCCA ERROR
02965	UNDER CODE NOT FOUND NFCCA
02975	ERROR DELIVERY TO NFCCA CONVERSION
02985	CRB NFCCA OUTGOING CB2C ERROR
02995	CR NFCCA ERROR
03100	TRUNCATE TABLES OUT BASE2 ERROR
03101	THE BASE2 OUTGOING ERROR
03102	THE OUTGOING TABLES FORMATTING ERROR
03103	THE GENERATED FILE IS EMPTY
03104	ERROR READING FROM THE IPM_OUTGOING_COLLECTION TABLE
03105	ERROR READING TABLE IPM_OUTGOING_TABLE

Code	Message
03106	TRANSACTION TO CONTROL DOES NOT EXIST
03110	ERROR TRUNCATE TABLE IPM OUT
03115	ERROR READING CUTOFF_SEQUENCE
03125	ERROR DATA GENERATION OF CREDENTIALS
03135	BANK IS FOUND
03145	ERROR GENERATION OF HEADING
03155	ERROR GENERATION RETAIL AUTHENTICATING
03165	ERROR GENERATION OF TRAILER AUTHENTICATING
03175	ERROR Mette UPDATE THE CREDENTIALS TREATIES
03185	ERROR WHEN CLEANING THE TABLE MER_AUTHENTICATION_OUTGOING
03201	A HEADER RECORD ONLY PERMITTED BY FILE
03202	The ENÊTE CONTAINS ERRORS
03203	RECORDING END FILE NOT APPEAR BEFORE HEADING
03204	A RECORD ONLY PERMITTED BY END FILE FILE
03205	RECORDING END FILE CONTAINS ERRORS
03206	DETAIL RECORD CAN NOT BE BEFORE HEADING TREATY
03207	A DETAIL RECORD CAN NOT BE TREATY AFTER END RECORD FILE
03208	REGISTRATION DETAILS SEQ 02 CAN NOT BE FOUND MORE THAN ONCE IN DETAIL RECORD SEQ 01
03209	REGISTRATION DETAILS SEQ 03 CAN NOT BE FOUND MORE THAN ONCE IN DETAIL RECORD SEQ 01
03210	REGISTRATION DETAILS SEQ 04 CAN NOT BE FOUND MORE THAN ONCE IN DETAIL RECORD SEQ 01
03211	REGISTRATION DETAILS SEQ 05 CAN NOT BE FOUND MORE THAN ONCE IN DETAIL RECORD SEQ 01
03212	REGISTRATION DETAILS SEQ 06 CAN NOT BE FOUND MORE THAN ONCE IN DETAIL RECORD SEQ 01
03213	RECORDING END FILE WAS NOT TREATY
03214	ERROR THRESHOLD EXCEEDED
03215	HEADER FILE MISSING
03216	ERROR SEQUENCE NUMBER
03230	ABSENT REGISTRATION END FILE
03270	PARAMETER (%1) HAS WRONG VALUE (%2). POSSIBLE VALUES (%3)
03301	INVALID LAUNCH BATCH MODE

Code	Message
03302	LEGAL ID TYPE NOT DEFINED: 1%
03303	RECOVERY NOT ALLOWED . FOUND NO COUNTERS
03304	RECOVERY NOT ALLOWED . FILE PROCESSED
03305	RECORD IN WRONG POSITION . CURRENT: 1 % ; PREVIOUS: 2 %
04101	ERROR LOADING CB2C OF RECOVERY
04506	UPDATE BATCH STATUS : STATUS 1 % FOR UPDATE BATCH 2 % IS INVALID
04507	BATCH STATUS UPDATE : UPDATE THE STATUS OF BATCH IS IMPOSSIBLE 1 %
04508	STATUS UPDATE BATCH BATCH OF SELECTION PROBLEM WITH THE STATUS % 1 % 2
04519	READ ERROR BUFFER
04521	ERROR GETTING PARAMETERS FOR BATCH 1 %
04524	OPENING FILE ERROR
04527	CALL THE MAIN FUNCTION OF ERROR . PC
04550	BASE2 OLD DATE OF TREATMENT FOR BATCH 1 %
04551	BASE2 FILE ALREADY TREATED FOR BATCH 1 %
04552	PROBLEM WITH HEADER : 0 OR MORE HEADING FOR BATCH 1 %
04560	ERROR LOAD TABLE FROM THE BUFFER TABLE BASE2_INCOMING
04561	ERROR LOADING TABLES VSS
04577	ERROR TRUNCATE TABLE IN BASE2
04591	ERROR FILE BACK IS NOT TREATED
04596	ERROR PROCESSING WASTE VISA
04597	NOT REJECT THE HEADER OR FOOT FILE
04598	ERROR REMOVAL OF TRANSACTIONS TO BASE2_INCOMING_TAB_RECYCLE BASE2_INCOMING_TABLE
04599	ERROR DIVISION OF TABLES TABLE BASE2_INCOMING TO DIFFERENT LOADING
04688	ERROR TRANSACTION PROCESSING REJECTED IPM
04689	ERROR PROCESSING DETAILS OF FINANCIAL POSITION
04690	ERROR FILE PROCESSING OF MONEY SUMMARY
04691	ERROR HANDLING FEES IPM 1740
04692	ERROR PROCESSING APPLICATION RECOVERY IPM
04693	ERROR PROCESSING REPRESENTATIONS IPM
04694	ERROR PROCESSING chargeback IPM
04695	ERROR CHECKING ALL PRESENTATIONS IPM
04696	ERROR PROCESSING INCOMING IPM 1240

Code	Message
04697	ERROR TREATMENT OF CROSS RATE INTEGRATION
04698	ERROR PROCESSING ELEMENTS IPM REJECTED
04699	ERROR DIVISION IPM_BUFFER TABLE TO TABLE IPM_INCOMING_TABLE
04705	INCORRECT FILE STRUCTURE
05324	ERROR TRUNCATE TABLE IN IPM
05400	FILE EMPTY
05608	ERROR FILE ARCHIVING
05609	ERROR CODE HEADER FORMAT
05610	ERROR REMOVING OLD HEADER
05611	FILE IS REJOINDER
05612	SEQUENCE FILE IS OLD
05613	FIELD OF REFERENCE IS INVALID HEADER
05614	ERROR FORMAT THE TRAILER
05615	THE TOTAL NUMBER OF REGISTRATION IS DIFFERENT FROM THAT CONTAINED IN THE ENQUEUE
05616	FIELD DEFAULT RECORD NOT FOUND IN THIS FILE
05617	ERROR ON RECORDING FORMAT RETAIL
05618	REGISTRATION DETAILS REJOINDER IN THE FILE
05619	THE REGULATORY CONTEXT IS INVALID
05620	MCC INVALID
05621	ERROR FILE CONTAINS NO RECORD HEADER
05622	ERROR FILE CONTAINS NO REGISTRATION ENQUEUE
05623	ERROR DURING LOADING FILE IN THE TABLES
05624	DATE OF HEADING IS INVALID
05625	CURRENCY IS INVALID
05626	MORE HEADINGS IN THE FILE
05701	BANK RECEIVED IN THE FILE IS INCORRECT
05703	INCORRECT DATE FILE
05705	SEQUENCE INVALID FILE
05706	RECORD NUMBER OF INCORRECT
05800	HANDING LOGIN DOES NOT EXIST
05802	EN-QUEUE FILE MISSING
05805	DISCOUNT HEADER MISSING
05806	ENQUEUE SAVE MISSING
05808	TOTAL TRANSACTION FLOW INCORRECT
05811	TOTAL TRANSACTION FLOW INCORRECT

Code	Message
05814	HEADER FILE IN DOUBLE
05815	INVALID FILE HEADER
05816	IN DOUBLE TAIL FILE
05817	INVALID MESSAGE FORMAT
05831	ENQUEUE INVALID FILE
05840	ENQUEUE DISCOUNT INVALID
05841	DISCOUNT INVALID HEADER
05842	FILE REJECTED : RELEASE RATE THAN THE THRESHOLD
05843	DATE FILE IS GREATER THAN THE DATE TREATMENT
08033	ERROR RESET COUNTERS
08034	ERROR CALCULATION OF COUNTERS
08035	MISSING PART STANDARD
08036	MISSING PART EMV1
08037	MISSING PART EMV2
08038	ADDITIONAL DATA MISSING PART PFA
08039	EMV SUPPLEMENT SAFETY PART 1 MISSING
08040	ERROR CONTROL DISCOUNTS CANCELLED
08041	ERROR RECYCLING TRANSACTIONS BASF RAIL ROAD '
08042	RECYCLING OF TRANSACTION ERROR ' TELECOM '
08043	RECYCLING OF TRANSACTION ERROR 'FOOD '
08044	REGION FOUND
08045	THE REGION IS SELECTED AT RANDOM FROM THE REGION TABLE
08046	THE FORMAT OF THE DATA IS INVALID
08047	NBR . REC . WHEREAS THE BANK IS DIFFERENT NBR . RECEIVED
08048	NBR . REC . WHEREAS AGENCY IS DIFFERENT FROM NBR . RECEIVED
08049	ENREGISTRMENT BANK IS INVALID
08050	ENREGISTRMENT AGENCY IS INVALID
08051	MODIFICATION OF BIC WITH VALUE RECEIVED FIB
08052	REGISTRATION UNEXPECTED
08053	ADDRESS LINE IS EMPTY
08054	ZIP CODE NO DIGITAL
08055	FOUND COUNTRY
08056	POSTAL CODE NOT FOUND

Code	Message
08057	AT LEAST 3 LINES OF ADDRESSES MUST BE KNOWLEDGEABLE
08058	WRONG ADDRESS CREATED ON CEDEX
09001	THE SEQUENCE LENGTH VACATION IS OVER 11
09002	THE SEQUENCE LENGTH IS GREATER THAN 18 CRE
09003	INVALID MODE GENERATION
09004	SEQUENCE OF CRE TO DUPLICATE IS INVALID
09005	ERROR WHEN CALLING STORED PROCEDURE
09006	LIST OF BANKS CONTAINS AT LEAST AN UNKNOWN BANK
09007	SEQUENCE IS NOT REQUESTED OR MORE IN THIS TABLE CRE
09008	SETTING THE NETWORK IS INVALID CRE
09009	THE TYPE SETTING THE RECORD IS INVALID
09010	ERROR DURING INITIALIZATION GRAPH , SEE LOG CLOVER
09999	TASK 1 % ENDED unsuccessfully
14551	FILE IS ALREADY TREATY
14552	HEADER FILE MISSING
15000	THE MONTHLY TASK MUST BE RUN BEFORE CREDIT DAILY TASK.
15001	BUSINESS DATE < LAST PROCESSING DATE OF CUTOFF DAILY : %1
15002	BUSINESS DATE < LAST PROCESSING DATE OF CUTOFF MONTHLY : %1
15003	WRONG VALUE OF TRANSACTION AMOUNT or TRANSACTION CURRENCY : %1
15006	%1 IS NULL
15007	WARNING ! THE NB OF INSTALLMENT IS TOO BIG.
15008	%1 IS NEGATIF
15009	GLOBAL VARIABLE %1 MISSING
15010	CROSS DEPENDENCIES BETWEEN PARAMETERS OF CALCULATION TERM DUE/OVERDUE DATE AND AUTO DEBIT DATE : %1
15011	ERROR VALUE OF BASE TERM DUE DATE : %1
15013	ERROR VALUE OF BASE OVERDUE DATE : %1
15015	ERROR VALUE OF BASE AUTO DEBIT DATE : %1
15017	ERROR VALUE OF BASE DEBIT VALUE DATE : %1
15019	ERROR VALUE OF BASE DUE DATE : %1
15020	V_BASE_VALUE_DATE IS WRONG
15023	CURRENCY OF LOAN TRX # THAN CURRENCY OF SHADOW ACCOUNT : %1

Code	Message
15024	WRONG DATA (DEBIT/CREDIT) RESULT
15026	THE INDEX FEES ON EVENT IS LINKED TO THE PROFILE BUT NO RELATED FEES DECLARED IN %2 FOR %1
15027	THE INDEX FEES ON TRANS IS LINKED TO THE PROFILE BUT NO RELATED FEES DECLARED IN %2 FOR %1
15028	INTERNAL ERROR EVENT ONE N YEAR BUT PERIOD CHECK IS <> THAN BASED ON REFERENCE DATE
15029	INTERNAL ERROR EVENT NEW ACCOUNT BUT PERIOD CHECK IS <> THAN BASED ON REFERENCE DATE
15031	ONE (OR MORE) OF MANDATORY ARGUMENT(S) IS (ARE) NULL : %1
15032	ACCOUNTS LINK CURRENCY: %1 IS DIFFERENT THAN CHAIN ACCOUNT CURRENCY: %2
15034	V_BASE_DEBIT_DATE IS WRONG
15035	V_BASE_DEBIT_VALUE_DATE IS WRONG
15037	V_BASE_GEN_DEPOSIT_DATE IS WRONG
15039	PROCESSING STEP <> TRX_WAIT_DAILY
15040	INTERNAL ERROR: DIFFERENCE BETWEEN SHADOW_ACCOUNT_NBR TRANSACTION
15042	INTERNAL ERROR UNKNOWN BALANCE TYPE
15043	AUT BILL CURRENCY <> TRX BILL CURRENCY : %1
15045	ERROR IN SELECT FOR UPDATE %1 NOWAIT
16000	MANY PIN REQUEST FOR THE SAME CARD
16001	NO PIN REQUEST FOR THIS CARD
16002	THIS CARD IS NOT UNIQUE
16003	REASON CODE IS EMPTY IN CARD_REPLACE TABLE %1
16004	THE FEES FLAG IN LINKED PROD PRODUCT TABLE IS NOT THE SAME FOR THIS AUXILIARY PRODUCT %1
16005	FEES FLAG IS GROUPED ONE OR ALL THE FOLLOWING FIELDS HAVE NOT CORRECT VALUE %1
16006	PURGE DATES %1 ARE NULL FOR %2
16007	PRODUCT TYPE INCOHERENT
17008	UNKNOWN MIN DUE SCHEDULE CRITERIA VALUE
17014	INVALID AMOUNT %1
17015	BUSINESS DATE < LAST PROCESSING DATE OF CUTOFF MONTHLY : %1
18000	N/A
18001	N/A

Code	Message
19000	THE NUMBER OF RECORDS IS NOT THAT MENTIONED IN THE FILE TRAILER
19001	ERROR! THE ORDER NUMBER OF THE RECORD IN POSITION %1 PLEASE CHECK THE ORDER NUMBER OF THE RECORDS IN THE FILE.
19002	ERROR! THE FILE HEADER AND TRAILER ARE NOT COMPATIBLE. %1
19003	ERROR! THE SUM OF THE PAID AMOUNTS IS DIFFERENT FROM THAT MENTIONED IN THE FILE TRAILER. %1
99998	N/A
99999	ORACLE ERROR :% 1
R0702	THE REMITTANCE DOES NOT EXIST
R0703	ERROR IN THE CREATION OF REGISTRATION
R0704	CANCELLATION OF SURRENDER IS ALREADY CAPTURED
R0705	DISCOUNT ALREADY CANCELLED
R0706	THE REASON CODE IS INVALID
R0707	ERROR DURING THE TREATMENT OF CANCELLATION
R0901	CANCELLATION DATE IS GREATER THAN THE DATE BUSINESS
R0977	DELIVERY CONTAINS TRANSACTION REJECTED
T0601	TRANSACTION HAS ALREADY BEEN ANNUALEE
T0608	UNWANTED PROBLEM DURING THE SAFETY OF CANCELLATION
T0609	CANCELLATION CAUGHT WITH SUCCESS
T0671	CANCELLATION REMOVED SUCCESSFULLY
T0672	NO REGISTRATION DELETE
T0673	CANCELLATION REMOVED SUCCESSFULLY
T0674	NO REGISTRATION DELETE
T0675	CANCELLATION REMOVED SUCCESSFULLY
T0676	NO REGISTRATION DELETE
T0902	THE TRANSACTION TO UNPAID
T0903	CANCELLATION ALREADY CAPTURE
T0904	CANCELLATIONS ARE ONLY TOTAL FOR CB AUTHORISEE NETWORK
T0905	REASON FOR CANCELLATION IS INVALID
T0906	CANCELLATION DATE IS GREATER THAN THE DATE BUSINESS

6 Annexe C Encryption & decryption process

Here are some print screens of the encryption and decryption process:

Automatic encryption process of a report:

```

0208 132129979 00001 21233854 00001|5| *****
145231948 12517518 00000001|0| -----
1208 145231948 12517518 00000001|0| ----- [ Processed : 0000000000 ] -----
1208 145231948 12517518 00000001|0| -----
1208 145231948 12517518 00000001|0| End LoadBatchNewCounters()
1208 145231948 12517518 00000001|5| End Execution PL/SQL OK
1208 145231948 12517518 00000001|5| openssl smime -encrypt -aes256 -in /pwrcard/
everissd/usr/data/MERCHANT_TURNOVER_FILE_201408
1208 145231997 12517518 00000001|5| Start Archiving File
1208 145232009 12517518 00000001|5| End Archiving File OK
1208 145232009 12517518 00000001|5| Start Execution PL/SQL
1208 145232027 12517518 00000001|5| End Execution PL/SQL OK
1208 145232028 12517518 00000001|5| Start PutPcardErrors()
1208 145232028 12517518 00000001|5| END PutPcardErrors(OK)
1208 145232028 12517518 00000001|5| Start LogPcardErrors()
1208 145232028 12517518 00000001|5| ***** START PCARD_ERRORS
*****
1208 145232029 12517518 00000001|5| ***** END PCARD_ERRORS
*****

```

7 Document history

Document version	Status	Date	Modification	
1.0	Release	07/05/2020	Release	
1.1	Release	15/07/2020	Review of the Running a batch section	